Evidence-Monotone Audio Representation: Preventing Provenance Promotion Across Derived Audio Transformations

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Abstract

Derived audio can keep its waveform while strengthening what it says about its origin. Under a constructed boundary-only inheritance policy, a trimmed and re-encoded clip of captured and generated material emerges described as captured-only. We specify an operator contract for derived audio: over the complete required source set, provenance unions, same-channel support minimises, applicability intersects and lineage unions, and an unverified source interval is absorbing. The algebra is a known meet-semilattice construction; new are its executable temporal instantiation with declared kernel footprints, a signed C2PA transport, a consumer verifier and their evaluation. The contract prevents promotion relative to the declared dependency map; whether that map matches actual processing needs separate justification. Complete-source propagation failed none of 984 213 contract checks and promoted in no interval over 10 000 adversarial timelines, 600 mixed-origin clips and 600 PartialSpoof and 600 PartialEdit third-party utterances processed by stock FFmpeg, where boundary-only inheritance promoted in 4 200, 4 158 and 4 315 of 4 800 runs. A holdout challenge showed that the measured MP3 and time-stretch declarations are not universal. A strict profile assigns whole-asset dependency to measured declarations: safe by construction, it promoted nowhere and dilutes those operators' claims almost entirely. Consumer tests checked assertion consistency against the declarations; an end-to-end signed workflow accepted all 20 honest trim derivations and rejected all 20 promoted variants; signing left decoded audio unchanged. Independent runs reproduced the conformance, promotion and oracle outcomes of an earlier release. The contract does not detect deepfakes, and a valid signature proves only that an assertion was signed.

Keywords: digital evidence, media provenance, content credentials, C2PA, audio forensics, derived media, reproducibility

1. Introduction

Speech can now be generated convincingly enough that "is this recording real?" is answered less and less reliably by listening. Detection asks whether a signal carries the traces of synthesis (Pham et al., 2025; Wang et al., 2026); provenance asks what an asset's own signed history says about where it came from (Coalition for Content Provenance and Authenticity, 2026a). This work is about the second question, so nothing here is a detection result.

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Provenance systems bind assertions to an asset. They do not, on their own, say what a *derived* asset is entitled to assert, and derivation is routine: a clip is trimmed, spliced, resampled and re-encoded before it reaches an examiner, and each step writes a new manifest. The C2PA specification names the relationship of such an asset to its input `parentOf` (Coalition for Content Provenance and Authenticity, 2026a, Table 10, §18.16.3), and its vocabulary can carry everything the problem needs: `componentOf` ingredients with their own histories, a `digitalSourceType` for generative origin, assertions scoped to a temporal range. What the text does not prescribe, and we searched it for such a rule, is the *reduction*: the aggregate evidence state a derived interval should emit when the component histories it represents disagree. The coalition's July 2026 guidance on identifying synthetic content develops the vocabulary for declaring AI involvement and localising it to regions (Coalition for Content Provenance and Authenticity, 2026b), which is how to *represent* what happened to part of an asset, not what a derived interval may *claim* once its parts disagree.

Consider five consecutive source intervals of a recording whose third is generated speech. A derived asset representing all five can be described from its retained boundaries, which are captured, and emerge labelled captured-only; the decoded audio is bit-identical either way (Figure 1). An examiner reading that claim is being told something stronger than the sources support, by a chain in which no signature was forged, and wherever a disclosure obligation is discharged by reading a manifest rather than inspecting a signal the result is a confident wrong answer with a valid signature attached.

We call this *provenance promotion*: a derived representation acquiring an evidential claim its complete source material does not support, without any change to the decoded signal. The claim may be strictly stronger, such as captured-only for mixed sources, or incomparable, such as captured for a generated source; both are promotion. The requirement that prevents it, *evidence monotonicity*, is that along a chain of representation-only transformations the emitted claim is no stronger than the meet of every required source (the strongest claim all of them support), ancestry accumulates rather than disappears, and the absence of verified evidence is absorbing.

Digital forensics already treats the step from stored evidence to software-derived representation as a place where meaning can change, and two failure modes recur: what a representation aggregates away (Hargreaves et al., 2024; Andersen et al., 2025), and how far the same nominal input is read differently by different builds and versions (Nordvik and Axelsson, 2022; Külper et al., 2026). Derived-media provenance meets both at once: heterogeneous source evidence has to be reduced to one claim, and the processing dependencies that decide which sources count are fixed by the build, so a conservative reduction rule and validated dependency declarations are both needed.

An earlier instantiation of that requirement exists for spatial data, released as open software, where derived terrain geometry can overstate the evidence of the source arc it replaces (Urias, 2026). This work asks whether the requirement is usable in a temporal medium with a real signed transport and real processing software, and what it costs. Only the algebra transfers; no earlier result is reused, and Supplementary Note S1 lists the inheritance row by row.

The algebra, a meet over a semilattice of labels with union of ancestry and an absorbing bottom element, is an instance of constructions that are decades old, in semiring-based provenance annotation (Green et al., 2007) and in label lattices for secure information flow (Denning, 1976; Goguen and Meseguer, 1982). We do not claim the non-amplification property as a new theorem; we state it as an instance, prove the footprint-monotonicity result the temporal instantiation relies on, and locate the contribution in what is executable and measured.

The propagation rule operates on evidence intervals indexed in decoded-sample coordinates, so

the interval representation is independent of the audio container. A transformation that preserves decoded samples and their alignment, such as lossless transcoding between WAV and FLAC, takes a zero kernel footprint (the reach in neighbouring source samples that an operator can mix into one output sample); other transformations need their own dependency model. MP3 appears often in what follows because, as a lossy and adaptive encoder, it is the hardest case to declare, not because the method is limited to it.

A signature check and this contract test different things. Signature validation establishes the integrity of an assertion; evidence consistency checks whether the assertion follows from the declared sources. C2PA already records ingredient relationships, and the contribution here is the executable reduction over them and its evaluation.

Contributions. Each is traceable to a completed experiment.

1. **The contract, and its temporal instantiation.** We state an operator contract for derived media without reference to any modality, and present its non-amplification property as an instance of a known meet-semilattice construction (Section 4). We instantiate it over sample-exact half-open intervals for eight representation operators, each with a declared kernel footprint, analytical or measured. A footprint-monotonicity result shows that over-approximating the required source set is always safe (Section 5, Proposition 2).
2. **A signed transport, and an ablation of the rule.** Interval evidence is carried inside C2PA temporal regions of interest and survives derivation and re-signing on 3 containers with 120 round-trips. A C2PA-native composition joins signed captured and generated `componentOf` ingredients (Sections 5.4, 7.6, 7.9). A policy ablation shows that dropping either complete-source aggregation or declared kernel footprints fails within the tested policy family (Section 7.3).
3. **Evidence that it holds, and what it costs.** An executable conformance suite runs 984 213 checks over every source word of length up to 8 over $\{C, G, \perp\}$, with 0 failures. A two-language differential over 10 860 frozen cases gives 0 disagreements; those frozen cases are the conformance target a third-party reimplementation can be run against (Sections 7.1, 7.12). 600 mixed-origin clips are processed entirely by stock FFmpeg, which validates the interval mapping against software we do not control (Section 7.4), and 600 PartialSpoof utterances and 600 PartialEdit files carry third-party ground truth (Section 7.5). A consumer verifier recomputes every claim from a signed dependency declaration (Section 7.11). Agreement on output length says nothing about how far a dependency reaches, so differential influence probes test the declared footprints separately (Section 7.7). Both passes are packaged as `tools/calibrate_footprint.py`. The declarations are then challenged on held-out contexts, amplitudes and positions, and the encoder and stretcher declarations fail (Section 7.8). Cost is measured on two axes. In runtime, bookkeeping is 0.152% of the processing it accompanies and the assertion grows at 333 B per evidence interval (Section 7.13). The *evidential* cost is a claim-dilution metric, the share of output whose claim the footprints weaken. It is priced in samples per ancestry boundary rather than in fractions of the asset, and compared with a whole-asset claim (Section 7.10).

Scope of the guarantee. The contract prevents evidence promotion relative to the declared dependency map. Whether that map corresponds to the actual processing needs separate justification, which analysis of the algorithm gives for some operators and calibration probes give only at the operating

points they tested. The strict profile applies whole-asset fallback to every measured dependency, within the paper's content-dependency model, where an operator's control dependencies are not counted.

The contract does not infer authenticity from acoustic features; it prevents a provenance-aware chain from promoting a mixed or unverified source interval into a stronger captured-only class, and Section 3 states what it does when evidence is absent. It is orthogonal to passive deepfake detectors and to watermarking, and composes with either. Transparency obligations under Article 50 of Regulation (EU) 2024/1689 became applicable on 2 August 2026 (European Parliament and Council of the European Union, 2024), which makes machine-readable origin claims about derived media operationally relevant; we make no legal claim about what any disclosure regime requires.

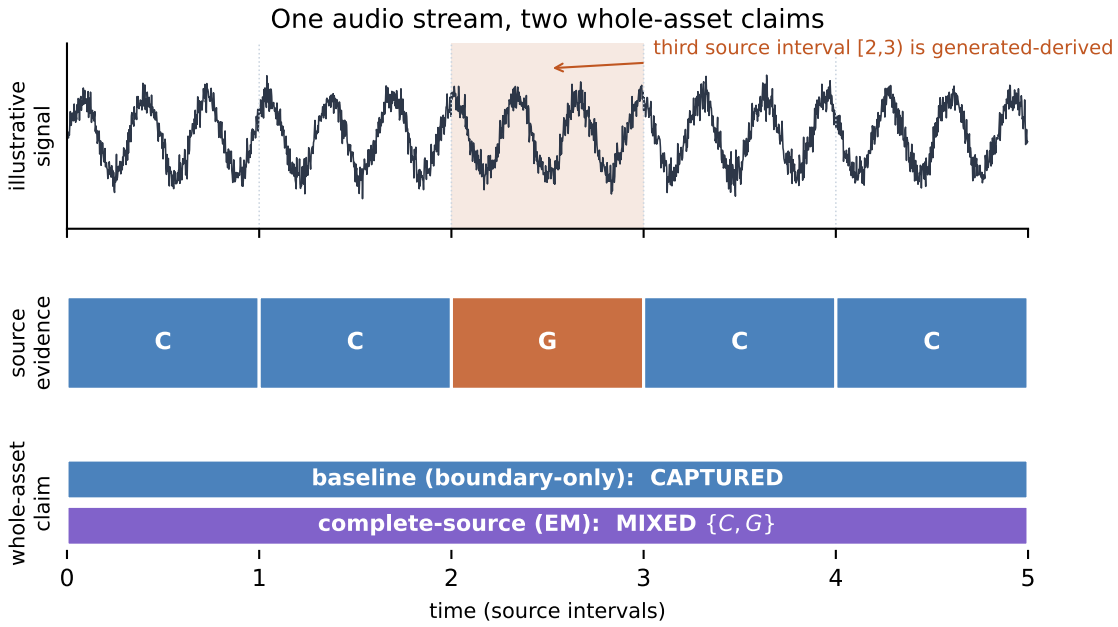


Figure 1: The minimal temporal counterexample. Five source intervals, of which the third is generated-derived. A boundary-only reading of the retained region sees captured evidence at both ends and may emit CAPTURED; the complete-source rule sees $C + C + G + C + C$ and emits MIXED $\{C, G\}$. The decoded audio is identical under both policies.

2. Related work

Detection of synthetic speech. Anti-spoofing and deepfake-speech detection classify a signal by its acoustic traces, and lose accuracy under unseen attacks and codecs, as documented (Wang et al., 2026; Pham et al., 2025). That line guarantees a decision about a signal; it does not constrain what a derived asset may assert about its own sources, and it needs no source evidence to exist.

Watermarking and soft bindings. Localised audio watermarking marks generated speech at synthesis time and detects the mark per region (San Roman et al., 2024); C2PA treats watermarks and fingerprints as *soft bindings* that recover a manifest separated from its asset (Coalition for Content

Provenance and Authenticity, 2026a, §18.10). Neither defines the aggregate claim of a derived asset whose sources differ, and a region carrying no mark is not thereby captured.

Cryptographic media provenance. C2PA binds signed assertions and transformation history to an asset, defines the ingredient relationships `parentOf`, `componentOf` and `inputTo`, and scopes assertions to temporal regions of interest (Coalition for Content Provenance and Authenticity, 2026a, §18.2.2.3, §18.16.3). Security analyses report problems in the cryptographic and trust layers (Golaszewski et al., 2026), and related work constructs assets whose manifest and watermark contradict each other while each passes its own verification (Nemecek et al., 2026). Our concern is a different layer, orthogonal to both: assuming keys are intact, signatures verify and no assertion is omitted in bad faith, is the assertion a conforming derived asset emits conservative with respect to every source interval it represents? The specification preserves a great deal, since a `componentOf` ingredient carries its own manifest and a generative component its `digitalSourceType` (Coalition for Content Provenance and Authenticity, 2026a, §18.16.3), and a consumer walking the full graph could reconstruct everything. What version 2.4 does not define is the aggregate: no normative rule states that a derived interval's own claim must be the conservative reduction over all its components rather than an inheritance from its parent. Our composition experiment is the evidence: every composition carries two ingredients whose declared source types disagree, and the official validator reports each as valid without computing or requiring any aggregate over them (Section 7.9).

Inferring audio provenance from the signal. Audio provenance analysis reconstructs content reuse and parent-child relations across a heterogeneous collection with no prior knowledge of its contents, combining partial matching with phylogeny (Gerhardt et al., 2024). That line infers a lineage where none was recorded; the contract here assumes signed source evidence exists and constrains how a known operator may aggregate it. A pipeline could reasonably use the first to recover a lineage and the second to decide what the recovered lineage entitles a derived interval to claim.

Provenance for edited audio. Chunk-localised speech provenance binds short segments to an issuer-signed original through perceptual fingerprints and Merkle commitments, addressing the fragility of hard bindings under re-encoding (Ono, 2026); zero-knowledge constructions prove an edit was drawn from an allowed set while hiding source, edits and device (Fu et al., 2026). Neither states an aggregation rule for when the represented sources disagree, and neither proves non-amplification.

Provenance models and information flow. W3C PROV records derivation as a graph (World Wide Web Consortium, 2013): a PROV graph for the five-interval counterexample would hold `wasDerivedFrom` edges to all five sources, a complete and true record, and still say nothing about which *claim* the derived segment may emit. The machinery for constraining that reading is established: provenance semirings annotate derived data so that combination is associative and distributes over alternative derivations (Green et al., 2007), though idempotence, which the meet here has, is not part of that framework and is incompatible with several of its instances; information-flow lattices require combination to move only in the conservative direction (Denning, 1976; Goguen and Meseguer, 1982). The property we need is the meet in such a structure, and we present it as such.

Forensic representation and tool-dependent interpretation. Hargreaves et al. (2024) model a forensic tool as connected processing stages in which an error introduced at one stage propagates downstream unless intermediate outputs are exposed and validated, and empirical work fills in what that costs. Andersen et al. (2025) reported examiner errors associated with misleading presentation in the

semi-automated tools used in that study, Cellebrite and APOLLO, naming aggregation, missing detail and timestamp presentation among the contributing factors. Interpretation also varies by implementation rather than by specification: Nordvik and Axelsson (2022) measured substantial inconsistencies in exFAT timestamp handling across the operating systems and the four forensic tools they examined, among them Autopsy and EnCase Examiner, and Külper et al. (2026) found largely undocumented changes in exported report output across twenty-five versions of Autopsy. None of these studies concerns provenance or demonstrates promotion, and the products named in them appear here only as their subjects. The premise they establish is that preserving source data is not sufficient if the derived representation changes the evidential meaning an examiner sees.

What is missing. Table 1 places these lines side by side. No row provides an operator-level complete-source rule instantiated over media time and carried in a signed transport. This work contributes that combination; the matrix behind the table, with search dates and query strings, is Supplementary Note S2.

Source attribution from the signal, such as identifying a recording microphone (Qamhan et al., 2023), infers origin from the audio, where this work reads no audio at all. A contract is useful only if its implementation can be checked, which is why the conformance suite of Section 7.1 is written against the requirements rather than against our code.

3. Threat model

In scope. Provenance promotion and laundering through representation-only transformations, under four conditions: source evidence exists; the processing system holds the source-to-output mapping; the operator is expected to conform to the contract; and cryptographic primitives and signing keys are intact.

Out of scope. Capture devices that lie about what they recorded; compromised signing keys; dishonest certificate authorities; unsigned media of unknown origin; speaker identity; the truthfulness of what is said; semantic misinformation; waveform-artefact deepfake detection; watermark removal; and full provenance stripping followed by construction of a fresh, dishonest trust chain. Attacks on the cryptographic and trust layers are the subject of separate analyses (Golaszewski et al., 2026) and are assumed away here precisely so that the residual semantic question is isolated.

Required behaviour when evidence is absent or invalid. The state is UNVERIFIED or INVALID, and the contract reads it as neither "fake" nor "captured". An asset can lose its manifest for benign reasons, and nothing in this work licenses reading a missing manifest as evidence of intent.

Malicious signer. Cryptographic validity proves that an assertion was signed, not that it is true; a signer willing to assert falsehoods can produce a valid manifest containing them, and no propagation rule can prevent that. The contract constrains conforming operators, not a dishonest issuer. A threat-model matrix pairing attacker capability against contract behaviour is Supplementary Note S3.

4. Formal model

4.1. Evidence records

Let a source asset be partitioned into elements. Each element x carries an evidence record

$$e_x = (P_x, S_x, A_x, L_x), \tag{1}$$

Table 1: Positioning. Each row names a guarantee the line already provides, then the evidence property it does not, by itself, enforce on a derived audio interval. The contribution here is the joint enforcement of the right-hand column through representation-only operators, not the reinvention of any left-hand guarantee.

Line of work	Established guarantee or finding	Left unenforced on a derived interval
Synthetic-speech detection (Wang et al., 2026; Pham et al., 2025)	A decision about a signal from its acoustic traces	What the derived asset may assert about its own sources
Localised watermarking and soft bindings (San Roman et al., 2024; Coalition for Content Provenance and Authenticity, 2026a)	Region-level presence of a mark; recovery of a separated manifest	Aggregation of heterogeneous source evidence into one claim
Cryptographic provenance (Coalition for Content Provenance and Authenticity, 2026a)	Signed assertions, ingredient relationships, temporal regions of interest	A rule requiring the derived claim to be conservative over all ingredients
C2PA security analysis and cross-layer attacks (Golaszewski et al., 2026; Nemecek et al., 2026)	Weaknesses in the trust, timestamp and validator layers; contradictions between layers	Behaviour of a chain in which every layer is intact and honest
Audio provenance analysis (Gerhardt et al., 2024)	Reconstruction of reuse and parent-child relations from unknown signals, via partial matching and phylogeny	What a derived interval may claim once its recorded evidence disagrees
Provenance for edited audio (Ono, 2026; Fu et al., 2026)	Chunk-to-original binding after transport; zero-knowledge edit correctness	An aggregation rule when represented sources disagree
Provenance graphs and lineage (World Wide Web Consortium, 2013)	A truthful record of derivation	Operator-level monotonicity as an executable contract
Provenance semirings and information-flow lattices (Green et al., 2007; Denning, 1976)	The algebra of conservative combination	A temporal media instantiation with a signed transport
Forensic tool representation and validation (Hargreaves et al., 2024; Andersen et al., 2025; Nordvik and Axelsson, 2022; Külper et al., 2026)	Evidence that representation and output can vary across processing stages, tools and versions	A conservative operator-level rule for what a derived interval may claim about heterogeneous source evidence
Verifiable chain-of-custody ledgers (Burri et al., 2020)	That a record of handling is complete and unaltered	What a derived asset may claim about material it represents

where P_x is provenance, S_x maps each semantically typed support channel to a value in the lifted domain

$$\mathcal{S} = \{\bot_\mu\} \cup [0,1], \qquad \bot_\mu \preceq s \ \text{ for every } s \in [0,1], \tag{2}$$

A_x records the scope over which each channel is declared applicable, and L_x is lineage. In the corpus, for example, an interval of LibriSpeech speech carries $P = \{C\}$, a high capture-support value applicable to the digital asset, and the identifier of its source recording as lineage; an interval of synthesised speech carries $P = \{G\}$ and a low value. A channel can also be *unavailable*, written $\perp_\mu$. We order it below every numeric value, so that a channel which disappears counts as a weakening; otherwise the rule "no emitted value is greater" would say nothing about it. Formally, $\preceq$ on $\mathcal{S}$ is the usual order on $[0, 1]$ with $\perp_\mu$ added as least element.

Provenance is a set of ancestry atoms drawn from $\{C, G\}$, for captured-derived and generated-derived sources. Mixed ancestry is $\{C, G\}$ and is produced only by union; it is never a primitive. Nothing in the algebra below depends on the size of this set: $\{C, G\}$ is the v1 instantiation, and a deployment carrying finer atoms, per capture device or per generator, inherits the propositions unchanged. The absence of verified evidence is a distinct state $\perp$. It is not an atom, and it is never silently mapped onto C or G. The labels CAPTURED, GENERATED, MIXED and UNVERIFIED are renderings; the normative fields are P, S, A and L, and REAL is not a state anywhere in the model, the implementation or the serialisation.

4.2. The claim order and its meet

Write $\widehat{\mathcal{C}} = \{\perp\} \cup (2^{\{C,G\}} \setminus \emptyset)$ for the set of provenance claims, ordered by

$$q_1 \sqsubseteq q_2 \iff q_1 = \perp \text{ or } (q_2 \neq \perp \text{ and } q_1 \supseteq q_2), \tag{3}$$

read "q_1 is no stronger than q_2": a larger atom set is a weaker claim and $\perp$, which asserts nothing, is the bottom element. It is a policy order: it governs permission to make an exclusive-origin claim, and is not an ordering of truth, credibility or forensic reliability; a known mixture is not the same as uncertainty about origin, and neither captured nor generated means true or false speech. This is a partial order under which two claims always have a greatest lower bound but need not have an upper bound ($\{C\}$ and $\{G\}$ have none), so the structure is a meet-semilattice rather than a lattice, with meet

$$q_1 \sqcap q_2 = \begin{cases} \perp & \text{if } q_1 = \perp \text{ or } q_2 = \perp, \\ q_1 \cup q_2 & \text{otherwise.} \end{cases} \tag{4}$$

This is a standard construction, an idempotent, commutative, associative combination with an absorbing element, as in provenance semirings (Green et al., 2007) and information-flow lattices (Denning, 1976); we record it rather than derive it.

4.3. The complete-source operator

Let a representation operator T produce output element y from the *complete required source set* $D_y = \{x_1, \ldots, x_k\}$. Two notions of dependency diverge for some operators. The *representational* dependency of y is the source content y stands for; the *computational* dependency is the set of source samples that numerically influence its value. The contract governs the representational reading, and the kernel footprint reconciles the two where signal processing makes them bleed together: a resampling filter makes neighbouring content influence y, and since that influence cannot be distinguished from representation at the contract's level of abstraction, the footprint conservatively enlarges D_y to cover it. Amplitude normalisation is the clearest case where the two readings diverge: each output sample represents one source sample, while the gain depends on the whole signal. Encoder bit allocation and stretch alignment are further control dependencies of the same kind (Section 9). Write $D_y^\mu = \{x \in D_y : \mu \in A_x\}$ for the sources that declare channel μ applicable. The contract requires:

(i) Provenance $P_y = \bigsqcap_{x \in D_y} P_x$, that is, the exact union of the required source provenance sets, with $\perp$ absorbing and with no invented ancestry.

(ii) Support if $P_y \neq \perp$, $D_y^\mu \neq \emptyset$, the emitted scope $A_y(\mu)$ of (iii) is non-empty and every member of D_y^μ supplies a value, then $S_y(\mu) = \min_{x \in D_y^\mu} S_x(\mu)$; otherwise the channel is *unavailable*; without the applicability condition a channel narrowed to an empty scope under (iii) would carry a number applicable nowhere. Requirement (iv) strengthens the condition $D_y^\mu \neq \emptyset$ stated here to $D_y^\mu = D_y$.

(iii) Applicability the *canonical* scope is $A_y^{\mathrm{can}}(\mu) = \bigcap_{x \in D_y^\mu} A_x(\mu)$, and the emitted scope must satisfy $A_y(\mu) \subseteq A_y^{\mathrm{can}}(\mu)$, never a broader one; an empty emitted scope makes the channel unavailable. Conformance is the containment; the canonical reduction, which the reference implementation emits, takes $A_y(\mu) = A_y^{\mathrm{can}}(\mu)$. Theorem 1 states separately what holds for conforming and for canonical operators.

(iv) No partial subsets a channel is available on the output only if every required source declares it applicable *and* supplies a value: $D_y^\mu = D_y$ is required, not merely $D_y^\mu \neq \emptyset$. A channel unreported by some required source, or not applicable to it at all, is unavailable and is not computed from the remaining sources. The second half makes Proposition 2 hold for channels: without it, enlarging D_y by a source that declares an otherwise inapplicable channel would raise that channel from unavailable to a value.

(iv′) Unverified absorption for channels if any required source is in state $\perp$, every numeric channel on the output is unavailable.

(v) Lineage $L_y = \bigcup_{x \in D_y} L_x$, identifying the complete required source set. The record is carried per evidence-homogeneous interval, so L_y names the assets required by *that* interval; it does not record where in those assets the interval's sources lie, which only the mapping that produced it determines.

Requirements (iv) and (iv′) are where this model is strictly stronger than its spatial predecessor. (iv) costs expressiveness: sources with different channel schemas yield no channel on a shared interval. (iv′) is its companion for $\perp$, since computing a channel from the remaining verified sources would be the partial-subset computation (iv) forbids. Both are tested separately.

4.4. Properties

The guarantee below is that a derived claim can only lose.

Proposition 1 (Non-promotion)**.** *Under (i)–(iv′): $P_y \sqsubseteq P_x$ for every $x \in D_y$, and no atom appears in P_y that appears in no required source; if any required source has $P_x = \perp$ then $P_y = \perp$ and no numeric channel is emitted; and for every channel μ, $S_y(\mu) \preceq S_x(\mu)$ for every $x \in D_y^\mu$ while $A_y(\mu)$ is no broader than any $A_x(\mu)$.*

Proof. By (4), P_y is $\perp$ if any source is, and $\perp \sqsubseteq q$ for every q; otherwise it is the union of the source sets, which contains each P_x and is therefore $\sqsubseteq P_x$ by (3), and membership of an atom in P_y requires membership in some P_x. Requirement (iv′) withholds every channel when $P_y = \perp$. A minimum over a set is bounded above by each member, (ii) refuses to emit unless every applicable source supplies a value, and repeated intersection in (iii) cannot enlarge a set. □

Theorem 1 (Closure under composition)**.** *Let a finite chain of operators each satisfy (i)–(v), including (iv′).*

(a) The composed claim is no stronger than the meet over the union of all ancestral required source sets, the composed lineage is their union, every available composed channel value is bounded above by every applicable ancestor, and every composed scope is contained in the intersection of the ancestral scopes.

(b) If every operator in the chain emits the canonical reduction, the composition equals the canonical reduction over the union of the ancestral sets.

Proof. Associativity and commutativity of $\sqcap$ make the composed provenance the meet over the union of the ancestral sets, so no ancestor is dropped or invented; absorption of $\bot$ is preserved inductively because no later operator receives a verified element in place of an unverified ancestor. For a channel μ, each operator either refuses the channel or emits a value bounded by its applicable inputs, and transitivity of $\preceq$ bounds the final value by every applicable ancestor. Where the channel stays available at every step, each step emits exactly the minimum over its inputs, each input is itself the minimum over its own ancestors, and a minimum of minima over sets whose union is the ancestral set is the minimum over the whole set; the final value therefore equals that minimum, not merely bounded by it. Repeated intersection cannot broaden a scope, and lineage composes by union of the per-operator source maps; this gives (a). For (b), canonical operators never narrow a scope below the intersection, so a channel is available after the chain exactly when it is available over the ancestral union, and the minima and intersections above become equalities. Part (b) does not extend to operators that narrow a scope as (iii) permits: an operator that narrows a channel's scope to $\emptyset$ makes it unavailable to every later step, while the canonical reduction over the original ancestor would keep it. That outcome is conservative and satisfies (a), but it is not the flattened canonical reduction. □

Theorem 1 is the standard consequence of combining in a meet-semilattice. The next proposition is not specific to time, since its proof uses only inclusion, union, minimum and intersection, but it makes the temporal instantiation implementable against real processing software: more possible source dependencies may weaken an output claim and can never strengthen it, so erring towards a larger footprint errs safely.

Proposition 2 (Footprint monotonicity)**.** *For finite source sets $\emptyset \neq D_y \subseteq D'_y$, the claim the canonical reduction computes from D'_y is no stronger than the claim it computes from D_y, and for every channel μ the value computed from D'_y is $\preceq$ the value computed from D_y in $\mathcal{S}$.*

Proof. Enlarging the index set of a union can only add atoms, and by (3) a superset is a weaker claim; adding a source in state $\bot$ sends the claim to $\bot$, which is weaker still.

For channels the argument runs through (iv), which requires $D_y^\mu = D_y$ for availability. Suppose μ is available on D'_y. Then every member of D'_y declares μ applicable and supplies a value, so every member of $D_y \subseteq D'_y$ does too, and μ is available on D_y as well; the emitted value is a minimum over the larger set D'_y, which is no larger than the minimum over D_y. Suppose instead μ is unavailable on D_y. Some member of D_y then fails to declare μ applicable, fails to supply a value, or is in state $\bot$, or the scopes over D_y already have an empty intersection; that member lies in D'_y, and an intersection over more scopes is no larger, so μ is unavailable on D'_y as well, and $\bot_\mu \preceq \bot_\mu$. The remaining case is an added source whose scope empties the intersection in (iii): requirements (ii)

and (iii) then withhold the channel, giving $\perp_\mu$, which by (2) is $\preceq$ every element of $[0, 1]$. Every case therefore moves down $\mathcal{S}$.

The condition $D_y^\mu = D_y$ carries the weight here. Under the weaker $D_y^\mu \neq \emptyset$ the statement is false: with $D_y = \{x_1\}$ and μ not applicable to x_1, enlarging to $D'_y = \{x_1, x_2\}$ where x_2 declares μ with value v would raise the channel from $\perp_\mu$ to v. The hypothesis $D_y \neq \emptyset$ is needed as well: an empty set reduces to $\perp$, and enlarging it to one captured source would strengthen the claim. An output position with no required source is therefore not covered by the proposition, and the implementation marks it unverified. □

Proposition 2 is the practical result of this section: an implementation does not need the exact dependency set of a filtered operator, because a *conservative over-approximation* is always safe and only under-approximation can promote. That converts knowing which input samples every output sample depends on inside a third-party codec into a tractable requirement: declare a set that contains the true one. An analytical radius can establish that; a probe can only test it, and Section 5 says which of the two each operator's declaration rests on.

Signal transparency (P8). The contract must not change the audio. The canonical essence of an asset is the SHA-256 of its decoded stream rendered to signed 16-bit little-endian PCM at the asset's own rate and channel count, and it must be identical whether the EM assertion or the baseline assertion is attached. Whole-file hashes are unusable here because embedding a manifest changes the file by construction (Section 7.6). Equality of this canonical 16-bit rendering is established; equality of higher-precision decoded representations is not tested.

5. Audio instantiation

5.1. Temporal intervals

All positions are sample indices on a named asset and all intervals are half-open $[a, b)$, matching the C2PA convention that "All start times are inclusive of that moment in time, and all end times are, by default, exclusive of it" (Coalition for Content Provenance and Authenticity, 2026a, §18.2.2.3). A source timeline is a gapless ordered partition of an asset into evidence-homogeneous intervals.

The v1 support channel. Version 1 carries one typed channel, `capture-support`, applicable over the single scope `digital-asset`: the degree to which an interval's own record supports a capture claim, on $\mathcal{S}$. It is not a detector output and is not calibrated against any acoustic measurement: what this instantiation demonstrates is that a typed channel aggregates by minimum, absorbs, intersects its scope and survives the transport, not that its numbers carry forensic meaning. The corpus and transport experiments assign 0.90 to captured-derived and 0.10 to generated-derived intervals, so a mixed interval aggregates to 0.10; the enumeration sweeps a grid of values, and the universal statement over $[0, 1]$ comes from the proof. One scope cannot exhibit disjoint or overlapping scopes, nor a channel applicable to one source and not another, the case Proposition 2 turns on; a synthetic battery over 5 scopes covers those (Section 7.1).

An operator emits a set of *map pieces* (Figure 2): output range $[o_a, o_b)$ represents source range $[s_a, s_b)$ on a named asset, with a footprint radius in source samples. Where pieces overlap, as in a mix, the required source set of an output interval is the union over every covering piece. The emitted output partition is the pull-back of every source evidence boundary across every piece, which gives each emitted interval a stable complete required source set.

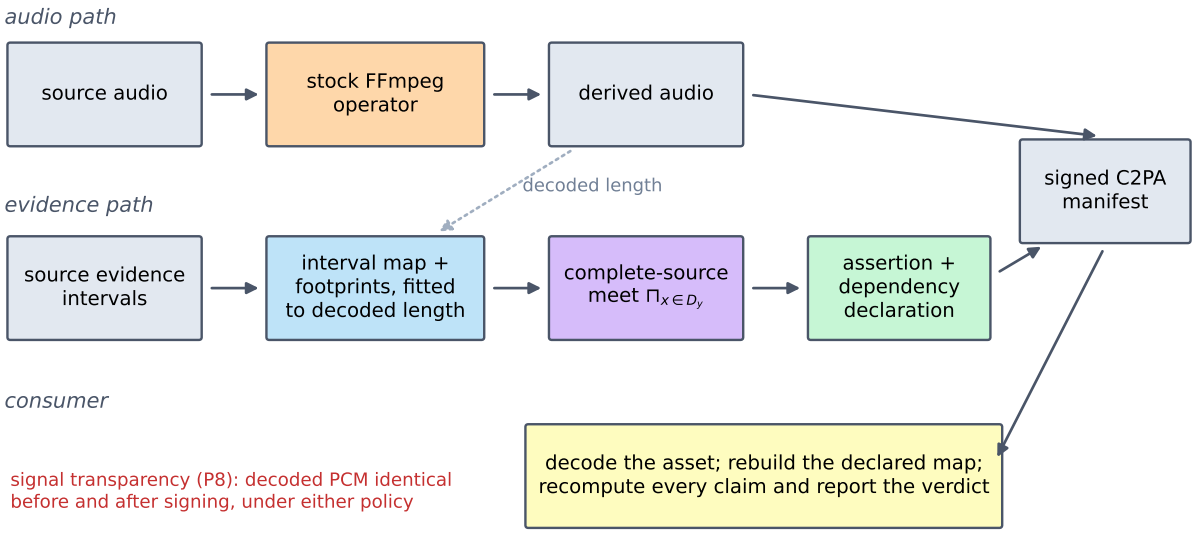


Figure 2: Where the contract sits. Audio and evidence travel separate paths that meet only at signing. Stock FFmpeg transforms the audio; the interval map, its footprints and the decoded output length determine each output interval's required source set; the complete-source meet produces the claims, which are signed with the dependency declaration. A consumer decodes the asset, rebuilds the declared map and recomputes every claim (Section 7.11). Signal transparency (P8) tests that signing preserved the decoded 16-bit PCM.

5.2. The v1 operator set and its kernel footprints

Table 3 gives the eight operators in scope with their mapping and declared footprint. Pitch shifting is not a further primitive: it composes a rate change with a compensating stretch, so Theorem 1 covers it and a named regression test exercises the composite. Denoising, separation and enhancement are out of scope for v1 because their dependency structure needs modelling of its own; generative editing differs more basically, since it may introduce new ancestry.

For a polyphase resampler the radius follows from the pinned filter length and rate ratio: 33 source samples here, which contains the largest reach measured in Experiment K (32) and in the holdout (32). The declared 97 adds a 64-sample margin chosen for frame alignment rather than measured; it only enlarges the footprint, so by Proposition 2 it cannot weaken the guarantee, and containment rests on the analytical radius alone. At 16 000 Hz `libmp3lame` emits MPEG-2 LSF frames of 576 samples; the radius takes the larger MPEG-1 long window, 1 152, plus the encoder delay, giving 1 728, and one frame of alignment guard. That derivation is necessary but not sufficient: the pinned `libmp3lame` runs with its bit reservoir enabled, which carries bits across frames, so the encoder is stateful in a way no window-and-delay argument captures. The declared 2 304 is therefore a *measured* declaration for this frozen configuration, containing a reach of 1 555 at the operating points probed in Section 7.7, and not a bound. For overlap-add time stretching the footprint covers the analysis window scaled by the tempo factor; for a frame-granular selector it covers the difference between the exact retained-sample model and the implementation's frame-aligned cuts. Two experiments check these numbers against software we do not control, and they bound different quantities: Section 7.4 measures temporal mapping deviation, which informs the margin, and Section 7.7 tests whether the *declared* footprint contains the measured source influence. For two operators the measured reach exceeds the analytical support and, at the probes of Section 7.7, is contained only by the declaration as a whole, which is why the declaration, not either component, is

Table 2: Footprint bases and the two evaluated profiles.

Basis	Operators here	Declared profile	Strict profile
Analytical	trim, concatenation, overlay, lossless transcoding, fixed-gain scaling, resampling	local	local
Measured	MP3 encoding, time stretching, retained-run selection	local, conditional	whole asset
Unpredicted output	tail beyond the modelled length	no claim	whole asset

what an implementation uses. An implementation on a build for which the two quantities have not been validated has no declaration to use, and the contract leaves three courses: refuse the operation, mark the interval unverified, which Section 4 makes absorbing, or widen to whole-asset dependency and pay the dilution of Section 7.10. Emitting the claim an unvalidated declaration would license is the one course forbidden.

Three kinds of footprint evidence, and two profiles. A declared footprint rests on one of three grounds, and the guarantee is as strong as its ground (Table 2). An *analytical* footprint follows from the algorithm and its pinned configuration. A *measured* footprint contains the dependency observed on a stated set of probes; probes can refute it and cannot establish it, because an adaptive operator re-decides when its input changes, and the holdout challenge of Section 7.8 refutes two such declarations on the reference build. The *whole-asset* fallback needs no measurement and gives up local specificity. The *declared profile* uses every footprint as declared; the *strict profile* keeps the analytical ones, replaces every measured footprint or margin by whole-asset dependency, and fits each emitted map to the length the processing actually decoded, giving any unpredicted tail whole-asset dependency. An implementation must never emit a local claim from a declaration that does not match its exact operator, build and configuration: it refuses the operation, marks the output unverified, or uses whole-asset dependency. Because whole-asset dependency contains any actual dependency, Theorem 1 applies to chains that mix analytical and strict steps, and such a chain cannot promote.

5.3. The baseline reference policy

The comparison policy throughout is *boundary-only, primary-parent inheritance*: the derived output inherits from its parent ingredient, evaluated at the first and last source elements of the retained region, ignoring kernel footprints. It is a **constructed reference policy**, not a description of any shipping product, and no claim is made here about what any named tool does. Its motivation is the specification text: a derived asset's relationship to its input is `parentOf`, "a derived asset or asset rendition of this ingredient" (Coalition for Content Provenance and Authenticity, 2026a, Table 10, §18.16.3); reading a parent's endpoints over a retained region is the baseline we define, an assumption of this paper rather than a reading the specification prescribes. Since the specification does not define the reduction, every implementation that derives an asset supplies one, deliberately or by default, and endpoint inheritance is one such rule. The baseline isolates one consequence of that underspecification without attributing it to any implementation; how deployed software behaves would need a survey this paper does not contain.

The baseline combines two independent shortcuts, boundary-only inheritance and blindness to the kernel footprint, so the experiments also evaluate the two intermediate policies that remove one shortcut each (Section 7.3). That ablation keeps the comparison from resting on the choice of baseline: a reader who disputes the composite policy can read the result off the intermediate arm.

Table 3: The v1 operator set. The analytical column is a support radius derived from the pinned algorithm configuration; the margin covers implementation behaviour that the analysis does not capture, such as frame-granular cut points and a correlation search range; the declared footprint is their sum and is the quantity Section 7.7 probes. All values are in source samples for the configurations used in the experiments, so transcoding takes a row per codec and the eight operators occupy nine rows; concatenation assembles the corpus clips rather than transforming them (Section 6.3), leaving the eight transformations Section 7 counts and, among those, the 7 single-source configurations Section 7.7 probes. By Proposition 2 erring large only weakens the emitted claim; erring small is a conformance failure. These values hold for one FFmpeg build and are not portable. The MP3 and time-stretch rows are measured, and the holdout of Section 7.8 exceeds both on the reference build; on the build of Section 7.14 the MP3 row is also under-declared for Experiment K's probes. Section 9 states the normalisation row's representational reading and its tested alternative.

Operator	Source mapping	Analytical	Margin	Declared	Basis
trim / crop	1:1 on the retained range	0	0	0	exact
concatenate	1:1 per part	0	0	0	exact
resample 16 to 8 kHz	$n_{\text{out}} \cdot f_{\text{in}} / f_{\text{out}}$	33	64	97	polyphase FIR
transcode (MP3)	1:1	1 728	576	2 304	MDCT window + delay; see note
transcode (FLAC)	1:1	0	0	0	lossless
amplitude normalisation	1:1	0	0	0	scalar gain; see note
time stretch 1.10	$t_{\text{src}} = \tau\, t_{\text{out}}$	529	2 048	2 577	overlap-add window
retained-run selection	explicit retained runs	0	4 096	4 096	frame-granular selector
mix / overlay	1:1 from every covering source	0	0	0	sample-wise sum

5.4. Signed transport

Evidence is carried as a namespaced C2PA assertion, `io.github.1d-coda.emaudio.evidence`, with one entry per output evidence interval. Each entry gives a temporal region of interest as an `npt` range (Coalition for Content Provenance and Authenticity, 2026a, §18.2.2.3), the sample-exact bounds alongside it because `npt` strings are lossy at sample resolution, the provenance atom list (`null` for unverified, never an atom), the available support channels, their applicability scopes, and the lineage set. The model allows real support values; the implementation stores finite binary64 values in $[0, 1]$ and rejects NaN, infinities and out-of-range values when a record is built. Support values are written as the shortest decimal that reads back to the same binary64 value, so a minimum survives serialisation exactly; a regression test fails if any rounding is reintroduced. Derivation records the source as an ingredient with relationship `parentOf`; the specification requires a corresponding `c2pa.opened` action carrying a hashed-URI reference to that ingredient assertion (Coalition for Content Provenance and Authenticity, 2026a, §15.11.3.2), which the claim generator inserts. Composition records each source as a `componentOf` ingredient carrying its full source manifest, with a `c2pa.placed` action referencing each component's ingredient assertion (Coalition for Content Provenance and Authenticity, 2026a, §18.16.3), so the transport exercises exactly the composition vocabulary whose missing aggregation rule motivates the paper (Section 7.9). The full schema, with a worked manifest, is Supplementary Note S4.

The transport is a bridge, not a re-implementation: signing and validation are performed by the official `c2patool`, so the transport guarantees are the tool's. The choice of C2PA is not load-bearing; the same assertion can be carried in a detached signed sidecar, and C2PA is used because it is the

deployed standard and tests the mapping against a real validator.

6. Methods

6.1. Study design

The experiments keep apart three questions that are easy to conflate. The first is mathematical: whether the implementation realises the stated guarantees. We answer it by exhaustive enumeration and by differential testing. The second is a mechanism question: whether a competing metadata rule produces counterexamples on adversarial and on realistic input. Frozen fixtures and a labelled corpus answer it. The third is an engineering question, what the contract costs, and timed repetitions answer it. Exact algebraic properties and finite-state classifications are reported as deterministic counts and are not assigned p-values. Table 9 states, for each class of claim, which evidence is used for it and what that evidence does not establish. Fixture failure frequencies characterise the frozen fixture family and are a **mechanism stress rate**: they are not, and must not be read as, an estimate of how often any deployed system promotes provenance.

6.2. Deterministic fixtures

Experiment A enumerates every source word of length 1 to 8 over $\{C, G, \perp\}$, giving 9 840 words, and pushes each through up to ten operator configurations. Two configurations require a minimum word length: the inner trim needs at least three elements and retained-run selection at least two. So the 3 single-element words receive eight configurations, the 9 two-element words nine, and the remaining 9 828 words the full ten, giving 98 385 cases. Each case is checked against the property suite and against a closed-form oracle that recomputes the expected state from the operator's nominal source ranges without the evidence algebra; zero-footprint operators must match it exactly, and kernel operators may only be weaker.

Experiment B freezes 10 000 timelines of 64 source intervals each, generated from a fixed seed, in which the majority of intervals are captured and one to four *non-boundary* intervals are replaced by generated or unverified evidence. Each is pushed through composition chains of depth 1 to 5 and, separately, through each operator alone. The anomalies sit away from the boundaries on purpose, so a policy that reads only the boundaries will miss them almost every time; this family measures the mechanism, not how often it occurs. A control arm places the anomalies uniformly at random instead. There the baseline promotes exactly when no anomaly lands on either end, which happens with probability $\binom{n-2}{k}/\binom{n}{k}$ for k anomalies of one type among n positions, so the measured rate can be checked against a closed form.

6.3. Mixed-origin audio corpus

600 clips are built deterministically, each of the form captured | generated | captured with sample-exact ground truth. Captured segments are crops of LibriSpeech `dev-clean` (Panayotov et al., 2015), CC BY 4.0; generated segments are produced locally by eSpeak NG formant synthesis from phrases written for this study. No speaker is identified and no voice is cloned. Segment durations are matched so that the generated region is a modest interior interval.

Given the same dependency map and source records, evidence reduction is independent of waveform content; adaptive processing can nevertheless change the dependencies the map must contain. A robustness arm tests the first part. It rebuilds 50 clips with a neural TTS (Piper, a VITS-architecture engine with a voice trained on the public-domain LJSpeech corpus). It also

overlays 50 clips with locally generated pink noise over the whole clip, mixed at a volume setting of −18 dB; this is a volume setting, not a measured signal-to-noise ratio.

Concatenation and every subsequent transformation are performed by **stock FFmpeg through its command line, with no project code in the signal path**: the artefact that does or does not exhibit promotion is produced by software the author does not control, and only the evidence policies are under study.

6.4. Metrics

We report promotion counts and rates, lineage omissions, unverified-to-verified promotions, support promotions, conformance failures, decoded-essence mismatches, manifest validation outcomes, model-versus-implementation sample deviations, runtime and metadata size. We do not report accuracy, precision, recall, F_1 or AUC: no classifier is evaluated. Timing is reported as medians with interquartile ranges over 30 repetitions on a single machine, an Apple M3 Max with 14 logical cores and 36 GiB, running macOS 26.5.2, Python 3.11.15, FFmpeg 9.0.1 and `c2patool` 0.27.2; every version is recorded in `results/PREFLIGHT.txt`, regenerated on each run. Each repetition processes 30 clips in one process with no affinity or power control and no warm-up, which moves the median by 0.09%.

6.5. Reproducibility and AI use

Every number in this manuscript is emitted by code into `results/machine_readable/`, converted into a LaTeX macro by a generator script, and used in the text through that macro; no result is typed by hand. Every figure is rendered from the same files. `run_all.sh` exits non-zero if any conformance check fails. Generative-AI tools were used as programming assistants and to improve language clarity, under the declaration at the end of this paper.

7. Results

7.1. Exhaustive conformance

Over 9 840 source words and 98 385 operator cases, the property suite and the closed-form oracle together evaluated 984 213 checks with 0 failures, including 363 explicit composition cases. Every named regression test passes, across the contract and review-regression suites together: 49 of 49, with 0 failures (Table 7).

That enumeration carries one channel over one scope, as the audio instantiation does, so it cannot reach the applicability cases of (iii). A separate synthetic battery covers those: over 5 scopes, chosen so that pairs are disjoint, nested and partially overlapping, it enumerates 441 source-set enlargements, skipping 184 that no valid source set realises, and checks each against Proposition 2, with 0 violations.

An earlier version of requirement (iv) withheld a channel only when a required source declared it applicable and reported no value, leaving a source to which the channel was not applicable at all to be silently skipped. Under that rule the enumeration still passed all 984 213 checks, because in v1 every non-$\perp$ source declares the one channel; the battery fails it in 80 of 441 cases, each an enlargement that raised a channel from unavailable to a value. The monotonicity check had the same blind spot, comparing a channel only when it was present on both sides. Both are fixed and the battery runs in the pipeline.

Passing checks are limited evidence. The superseded rule above, and the mutation tests that reject an inexact reduction, show that the suite detects these particular defects.

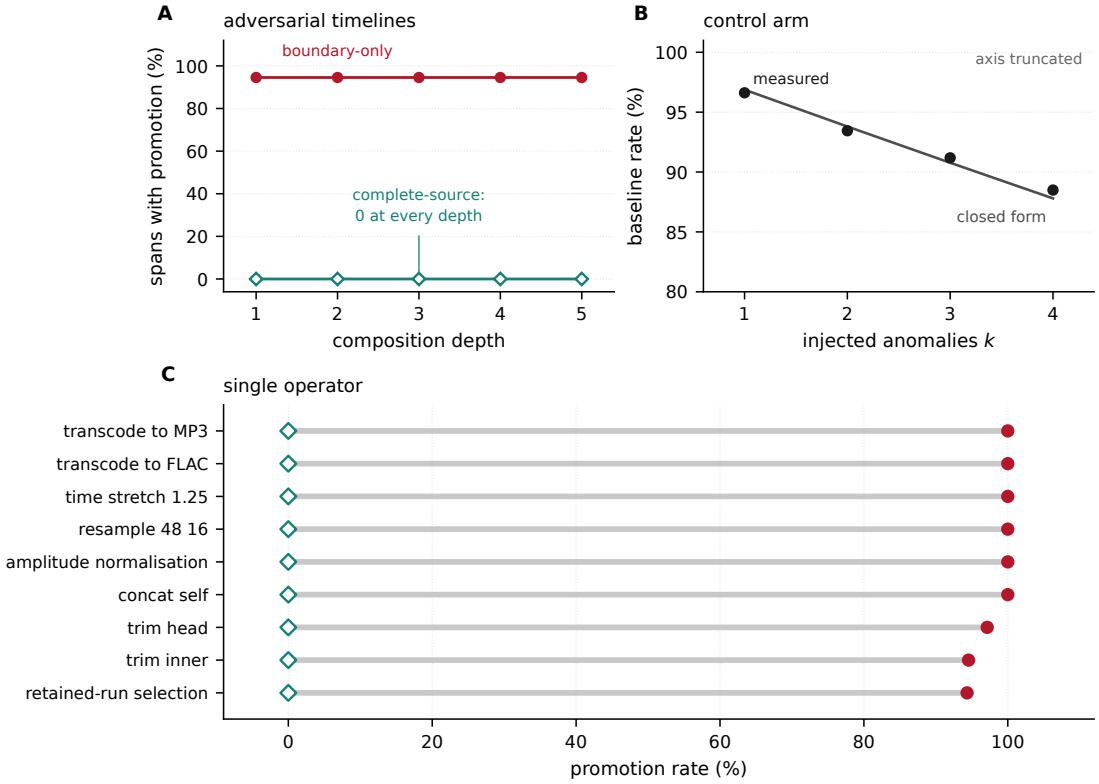


Figure 3: Boundary-only versus complete-source inheritance on 10 000 deterministic timelines. (A) Promotion does not grow with composition depth, because the loss is established at the first derivation and no downstream operator can recover evidence already discarded. (B) Control arm with anomaly positions uniform over all positions, measured against the closed form $\binom{n-2}{k}/\binom{n}{k}$, on an axis truncated to 80–100% to expose prediction error. (C) Each operator applied alone: filled circles are boundary-only, open diamonds complete-source, which is zero for every operator. Percentages throughout quantify mechanism stress on a frozen fixture family and are not real-world prevalence.

7.2. Adversarial timelines

On 10 000 frozen timelines, boundary-only inheritance promoted the provenance claim on 9 455 timelines (94.5%) at composition depth 1 and on 9 455 at depth 5; complete-source propagation produced 0 promotions summed over every depth, 0 unverified-to-verified promotions, 0 lineage omissions and no support promotion. Boundary-only inheritance omitted required lineage on 10 000 timelines at depth 1 and reported 7 222 unverified spans as verified.

The rate is constant in depth: the loss is established at the first derivation and is irreversible, because once the interior evidence is discarded no downstream operator can recover it (Figure 3A). Across the 9 operator configurations applied singly, baseline promotion ranged from 94.3% to 100.0% and complete-source propagation produced 0 promotions (Figure 3C). In the control arm, with anomaly positions uniform over all positions, measured baseline rates agreed with the closed-form prediction to within 0.0070 in absolute rate (0.70 percentage points) in every cell (Figure 3B). The adversarial arm's near-unity rate is thus a property of the construction, arithmetically predictable, and carries no information about deployed software; the informative quantity is the exact zero on the other side.

Table 4: Policy ablation over 10 000 timelines per arm. The interior arm places the anomaly inside the nominal represented range; the footprint arm places it outside that range but inside the declared kernel footprint. Each shortcut fails exactly one arm, and only the full contract clears both.

Policy	Inheritance	Kernel footprint	Interior anomaly	Footprint anomaly
B0	boundary-only	no	10 000 (100%)	10 000 (100%)
B1	boundary-only	yes	10 000 (100%)	0 (0%)
B2	complete-source	no	0 (0%)	10 000 (100%)
B3	complete-source	yes	0 (0%)	0 (0%)

7.3. Which shortcut matters: a policy ablation

The baseline combines two shortcuts, and removing them one at a time over 10 000 timelines per arm separates their effects (Table 4). In the *interior* arm the anomaly lies inside the nominal represented range, so only the inheritance axis can see it: both boundary-only policies promoted on every timeline (10 000 and 10 000) and both complete-source policies on none. In the *footprint* arm the anomaly lies just outside the nominal range but inside the declared kernel footprint: both footprint-blind policies promoted on every timeline (10 000 and 10 000) and both footprint-aware policies on none. Neither axis is redundant and neither is sufficient, which is the argument for specifying them together.

7.4. Mixed-origin audio through stock FFmpeg

On 600 clips, complete-source propagation recovered the evidence interval structure exactly, every boundary at sample resolution, for 600 clips, and propagated the supplied generated-source annotation with its exact bounds for 600, with no detector, no training and no acoustic feature. The ground truth is constructed rather than discovered: the question is whether evidence originating at the capture and generation steps *survives* a chain of representation-only transformations, not whether it could be inferred from the signal, and nothing here supports a claim that generated audio can be identified without evidence. The robustness arm behaves identically: 50 of 50 neural-TTS clips and 50 of 50 noise-overlaid clips recovered their full interval structure, with 0 and 0 promotions; under the noise overlay the captured segments correctly become MIXED while the generated segment stays GENERATED, since union adds no atom. Boundary-only inheritance promoted the whole-asset claim to captured-only on 600 clips and omitted required lineage on 600; the complete-source policy produced 0 promotions and 0 omissions.

Across 8 transformations and 4 800 runs (Figure 4; Table 7), boundary-only inheritance promoted on 3 600 runs and complete-source propagation on 0, with 0 lineage omissions. Those claims cover the modelled output length; in 888 outputs FFmpeg emitted more, 187 902 samples in all, and the declared profile makes no claim for them. Fitted to the decoded length, with whole-asset dependency on that tail, complete-source propagation under the strict profile promoted on 0 runs with 0 lineage omissions.

These counts score the claim over the whole output. Scored per interval, each claimed interval against the complete-source record for exactly its own output samples, boundary-only inheritance promoted on 4 200 of 4 800 runs (87.5%) and complete-source propagation on 0. The difference is overlay: its whole-output claim is a correct MIXED, because the generated source is a separate top-level ingredient, yet in 600 of its outputs the clip's own interval claims captured-only over samples whose sources include generated speech. Retained-run selection produced no promotion at either level in this corpus (0), because each retained run lay within a single evidence class, so its

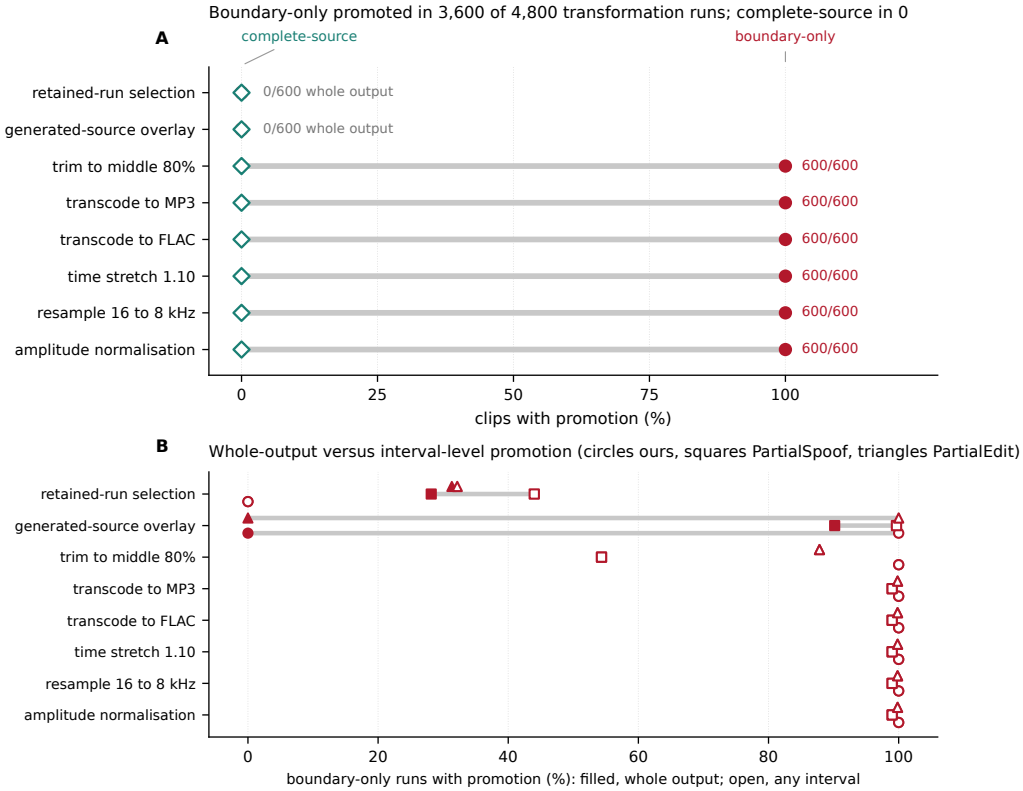


Figure 4: Promotion by transformation, 600 mixed-origin clips per transformation processed by stock FFmpeg, 4 800 runs in all. (A) Whole-output claim: filled circle, boundary-only reference policy; open diamond, complete-source propagation. Boundary-only promoted in 3 600 runs and complete-source in 0. (B) The same boundary-only runs scored per interval, on our corpus (circles), PartialSpoof (squares) and PartialEdit (triangles, Section 7.5); filled marks count whole-output promotion, open marks promotion in any interval. Overlay's whole-output claim is correct while one of its intervals promotes. Complete-source propagation promoted in no interval on any of the three corpora. The baseline is a constructed policy, and these are counts over frozen corpora, not prevalence estimates.

endpoints and its interior agree. A correct whole-output label can thus hide incorrect local claims: a correct whole-asset label does not establish the correctness of the claims attached to its individual intervals, and the interval-level count is the endpoint the contract is written for.

The interval model was validated against FFmpeg's actual output on every PCM run and on a 100-clip subset for the coded containers. The largest deviation between predicted and actual output sample count was 681 samples for retained-run selection, 471 for time stretching, 31 for MP3 and 0 for resampling; for all transformations the declared mapping margin covered it. That measurement compares output *lengths* only: it does not establish interior alignment, which the probes of Section 7.7 measure, nor the *support containment* Proposition 2 requires, because an operator can emit exactly the predicted number of samples while depending on sources outside the declared footprint; containment is tested directly in Section 7.7. Re-running each transformation and comparing decoded essence gave 0 mismatches, so the processing path is deterministic on this machine. The zero promotion count follows from Proposition 1 once the required source set is correct, and on overlay, during development, it did not; the falsifiable measurements are the mapping deviation and margin coverage here, and support containment in Section 7.7.

7.5. *Third-party mixed-origin speech: PartialSpoof and PartialEdit*

Our corpus places its own generated segments, so its ground truth is ours as well. PartialSpoof (Zhang et al., 2021a, 2023) embeds segments from the ASVspoof 2019 spoofing systems (Wang et al., 2020) inside bona fide utterances and publishes where they are, as timestamps (Zhang et al., 2021b). We took 600 spoofed development utterances: of the 15 847 spoofed development utterances in sorted identifier order, every 26th starting with the first, stopping at 600, a rule fixed in the fetch script before any result was computed, with the archive checksums published on the record verified by `tools/fetch_partialspoof.py`. Bona fide spans are captured and spoofed spans generated; a non-speech span takes the dataset's own 10 ms label. The same eight transformations ran unchanged.

The utterances have a median of 5 evidence intervals and last a median 3.45 s, of which a median 58.6% is generated: more fragmented and more generated than our clips. Two cautions apply. The dataset's timestamps are rounded to the nearest sample, and a non-speech span carries a 10 ms label, so boundaries are exact only to that resolution. The labels are annotations imported as source evidence, not signed capture or generation provenance, and the experiment propagates them: it is not a detection experiment, and the contract infers nothing from the waveform. The selected identifiers are listed in the result file.

Over 4 800 runs boundary-only inheritance promoted on 4 006 at whole-output level and on 4 158 (86.6%) at interval level; complete-source propagation promoted on 0 at either level (0 by interval), with 0 lineage omissions and 0 promotions under the strict profile. The consumer verifier of Section 7.11 accepted all 4 800 honest strict-profile assertions and flagged 4 006 of the 4 006 promoting runs, and 625 further outputs with another evidence-contract violation, a support value or lineage the sources do not give; checked against the decoded length, 4 800 of 4 800 strict-profile assertions verified. FFmpeg emitted more output than modelled in 1 137 outputs, which the fitted map covered with whole-asset dependency. The time-stretch mapping deviation reached 965 samples, inside its margin; overlay, whose margin is declared zero, deviated by 5 samples on one output, a mapping finding on third-party audio that the fitted map covers.

PartialEdit (Zhang et al., 2025, 2026) tests a newer generation of generator against the same rule: one to three words of a genuine VCTK recording (Yamagishi et al., 2019) are replaced by neural speech editing, a codec language model in its E1 subset, and the edited regions are published as timestamps. We took 600 of its 42 471 E1 files, the first 20 of each speaker, taking speakers in the order the archive presents them until 30 are complete, so that the download stops once the subset is complete; `tools/fetch_partialedit.py` verifies the label file's checksum and records each kept file's SHA-256. Edited regions are generated and the rest captured. The files have a median of 3 intervals and last a median 3.55 s, of which a median 19.0% is edited. Over 4 800 runs boundary-only inheritance promoted on 3 710 at whole-output level and 4 315 at interval level; complete-source propagation promoted on 0 (0 by interval) with 0 lineage omissions, and 0 under the strict profile, whose assertions verified against the decoded length in 4 800 of 4 800. The JavaScript verifier agreed with the Python one on all 2 400 cases it was given (0 disagreements). Three corpora from different groups and generator generations, our formant-synthesis splices, the 2019 spoofing systems and neural word editing, thus give the same outcome for the contract.

7.6. *Signed transport, derivation and provenance loss*

Across 120 signed round-trips over `wav`, `flac` and `mp3`, 40 clips per container, signing succeeded and validation reported the trusted state under the declared local anchor for 120 assets, with 0 validation failures. The EM assertion returned byte-equivalent from the validator for 120 assets.

After derivation with stock FFmpeg, re-signing with the parent as an ingredient, and re-validation, 120 derived assets validated and 120 recorded the `parentOf` relationship. AIFF holds the same 16-bit PCM as WAV and would take a zero footprint, but the `c2patool` release used here refuses to embed a manifest in it, so an AIFF asset would need a sidecar manifest, which this evaluation does not test.

Signal transparency holds exactly. Decoded essence was identical before signing, after signing with the complete-source assertion and after signing with the baseline assertion, with 0 mismatches against the pre-signing essence and 0 between policies. The whole-file hash changed in 120 of 120 cases, which is why the comparison is defined on decoded PCM; a clean-clone rerun makes the same point, every essence hash agreeing and no signed-file hash, each signature carrying its own timestamp.

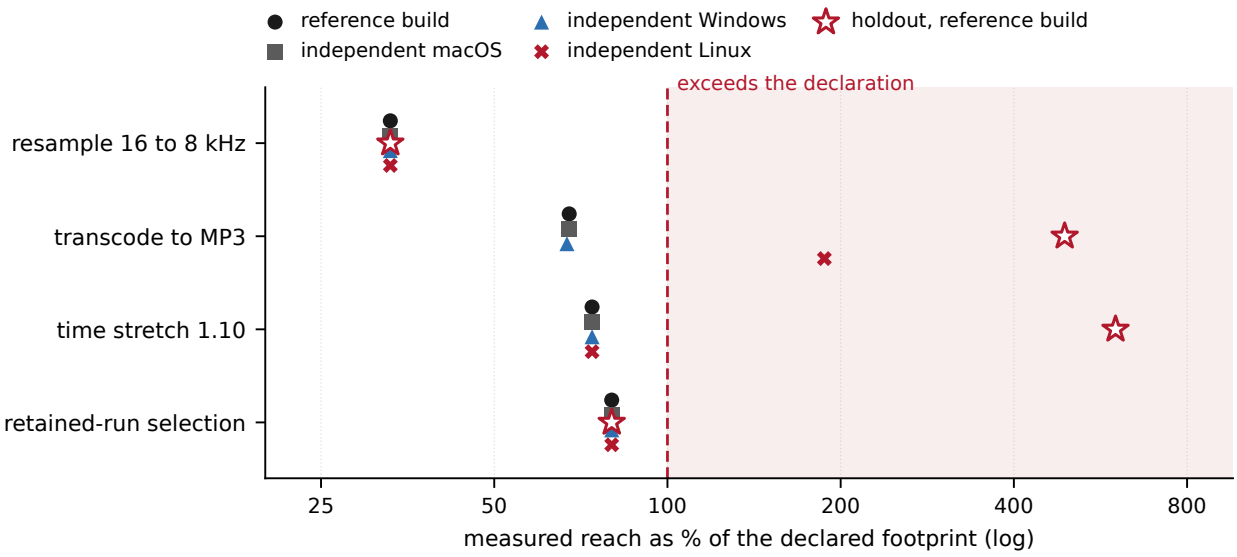
Under the five provenance-loss conditions over 60 clips, a valid manifest yielded MIXED in 60 cases, a valid derived manifest in 60, a removed manifest yielded UNVERIFIED in 60 cases, an asset modified after signing yielded INVALID in 60, and re-encoding without manifest preservation yielded UNVERIFIED in 60. No condition yielded CAPTURED, and there were 0 violations of the classification requirement. We do not claim that stripping is detected as malicious intent; the result is that the classification degrades in the conservative direction.

7.7. Kernel-support containment

Proposition 2 needs the declared required-source set to contain the actual dependency set, a statement about which source samples influence which output samples, not about how many output samples there are. We test it directly on the 7 single-source configurations; concatenation and overlay take several sources and have no temporal kernel to under-declare. The probe works on pairs of sources. For each configuration, each of 4 signal contexts and each of ten probe positions, we build two 16 384-sample sources that are identical except at one sample, where one carries an added impulse. Both go through the same stock FFmpeg command and both outputs are decoded. The entire difference between the paired outputs is therefore attributable to that one sample. Containment holds when the probe position lies inside the declared required-source range of every output sample whose value it changed, on 16-bit PCM at a threshold of one least significant bit.

Two design choices keep the probe honest about adaptive operators. The impulse sits on a non-silent carrier: on silence an encoder or stretcher spreads nothing, and its declaration would pass without being tested. The impulse is also scaled to each context's peak, so it nudges the operator's input instead of replacing it. The 4 contexts, all generated arithmetically (`experiments/support_containment.py`), are a steady low-frequency tone; the same tone at one part in a hundred, where a lossy encoder allocates fewest bits; sharp onsets with exponential decay, which drive an overlap-add stretcher hardest; and six inharmonic partials, where masking is most active. Lossy transcoding spreads a single source sample across up to 1 823 output samples at every position, resampling across up to 33, and overlap-add stretching is strongly position-dependent, reaching up to 2 023 at the 7 probes inside an overlap region and one sample elsewhere, which is why the test is run against the worst case rather than the median.

Across 280 probes, 52 005 output samples were influenced and 0 fell outside the declared support (Table 5; Figure 5); every context produced influence for every operator, so no row passes vacuously, and the source is long enough that the largest declared footprint does not span it. Retained-run selection first failed: a frame-granular selector emitted a frame beginning 3 277 source samples before a retained run's start, well beyond the margin declared at the time. That is an under-declaration,



reach in source samples; red: exceeds the declaration

	declared	reference	ind. macOS	ind. Windows	ind. Linux	holdout
resample 16 to 8 kHz	97	32	32	32	32	32
transcode to MP3	2,304	1,555	1,555	1,541	**4,317**	**11,292**
time stretch 1.10	2,577	1,906	1,906	1,906	1,906	**15,482**
retained-run selection	4,096	3,277	3,277	3,277	3,277	3,277

Figure 5: Kernel-support reach against the declared footprint. Each mark is the furthest any one source sample influenced the output beyond its nominal source range, as a percentage of the declaration on a logarithmic axis; the dashed line is the declaration, and marks to its right exceed it. Circles, squares, triangles and crosses are the containment probes of Section 7.7 on the reference build and the three independent runs (Section 7.14); stars are the maximum over the holdout challenge on the reference build (Section 7.8). The table gives each reach in source samples. Zero-footprint operators are omitted. Probes can refute a declaration and cannot establish one.

the one direction Proposition 2 does not forgive; the band was raised to 4 096, above the measured reach, and the operator now contains with a margin of 819. At these probes the bands are tight, which matters because a larger band dilutes more: resampling reached 32 against 97, MP3 1 555 against 2 304, and time stretching 1 906 against 2 577.

Under the model's own extent, 6 influenced samples, in retained-run selection and time stretching, lay beyond the map: output the operator emitted and the model did not predict. With the map fitted to the decoded length, those samples take whole-asset dependency (15 390 tail samples over all probes) and 0 influenced samples fall outside it; under the strict profile, 0. Interior alignment, the offset between where the model places a probed source sample and where its influence peaks, reached 205 samples for retained-run selection and 401 for time stretching, inside their declared margins, and 1 for resampling; a correct output length had not shown either.

7.8. Holdout challenge to the declarations

Passing those probes shows that each declaration contains what was measured at the operating points used while it was being set. A finite probe family can refute a declaration and cannot establish

Table 5: Kernel-support containment. Probed as Figure 5 describes. The measured reach is the furthest any influence extended beyond the nominal, footprint-free source range, so containment compares two concrete quantities rather than resting on an assertion: it holds exactly when the reach does not exceed the declaration. Headroom is the difference, so a small headroom means a tight declaration and a large one a conservative one. † Probed with a fixed gain: the content footprint of a sample-wise scaling. Estimating the gain from the input adds a whole-asset control dependency, measured in Section 7.8.

Operator	Declared	Measured reach	Influenced samples	Headroom	Outside
amplitude normalisation†	0	0	40	0	0
resample 16 to 8 kHz	97	32	1 131	65	0
retained-run selection	4 096	3 277	28	819	0
time stretch 1.10	2 577	1 906	10 971	671	0
transcode to FLAC	0	0	40	0	0
transcode to MP3	2 304	1 555	39 763	749	0
trim to the middle 80%	0	0	32	0	0

one for an adaptive operator. We therefore challenged the declarations with inputs that played no part in setting them, fixed before the run (`experiments/footprint_holdout.py`). The challenge used 6 signal contexts, 2 of them new (white noise and a speech-like voice). It used 4 absolute impulse amplitudes from 256 to 12 000, all within 16-bit range, and 8 positions disjoint from those above. This gives 192 probes per operator and 1 152 in total, on the reference build.

Resampling, retained-run selection and fixed-gain normalisation showed no exceedance, and resampling's reach stayed at 32 against its analytical 97. The two adaptive operators did not hold. MP3 exceeded its declaration in 4 of 192 probes, all on the low-level carrier at the largest amplitude, reaching 11 292 source samples; time stretching exceeded in 14, on the low-level carrier and on the held-out speech-like voice, at every amplitude from 256 upward, reaching 15 482. These long-range changes, largest for a large impulse on a quiet carrier, are consistent with adaptive bit allocation in the encoder and with the stretcher's correlation search re-aligning; the probes do not isolate either mechanism. Either way the change is a dependency of the output on the source sample, and a change of operating mode is no ground for excluding it. These two declarations therefore characterise the tested operating points on the build that produced them. Under the strict profile 0 probes of any content operator fell outside, which for the encoder and the stretcher follows from whole-asset dependency and Proposition 2; estimated-gain normalisation stays outside in 84 probes, a control dependency the profile does not count. Per-operator detail is in Supplementary Note S9. An exceedance is a real promotion when the source it misses carries weaker evidence. On the worst failing MP3 probe, labelling source sample 900 generated and every other sample captured, the declared map emits CAPTURED for output sample 7 355, whose true claim is MIXED; the strict profile emits MIXED. The worst time-stretch probe gives the same outcome at output sample 3 162.

Normalisation was run in two modes. With a fixed gain no exceedance occurred. With the gain estimated from each input, as peak normalisation does, 84 of 192 probes influenced output samples across the asset, up to 15 600 source samples away: a control dependency through the gain, not a content one. The contract labels origin by the content an output sample represents and so excludes control dependencies by definition; the measurement shows that this is a semantic choice, not an empirical property of the operator.

7.9. C2PA-native heterogeneous composition

The standards argument can be tested directly. For 30 fixtures, a captured source and a generated source are each signed with their own `digitalSourceType`, and stock FFmpeg concatenates them.

The composition is signed with both sources as `componentOf` ingredients carrying their full manifests. It also carries a `c2pa.placed` action referencing each source, and the EM assertion computed over the two component timelines. The composition is then derived across the ancestry boundary and re-signed with `parentOf`.

`c2pa.mixed` is defined for "multiple, previously placed (via `c2pa.placed` actions) audio ingredients (e.g., stems, vocals, drums, bass, etc) [that] are combined and optionally transformed" (Coalition for Content Provenance and Authenticity, 2026a, Table 8). Our fixture concatenates in sequence rather than summing, and we read that as placement rather than mixing, a *documented profile decision* rather than a normative one, since the specification settles neither that nor `c2pa.remixed` (Coalition for Content Provenance and Authenticity, 2026a, Table 8). `c2pa.mixed` presupposes `c2pa.placed`, so the placed actions are required under any reading, and the choice does not affect the contract, which reads the ingredient graph and the interval map, not the action label.

All 30 compositions validated as trusted under the declared anchor. Both `componentOf` ingredients were recorded in 30 cases, and both `c2pa.placed` references resolved in 30. The EM assertion returned byte-equivalent in 30. The whole-asset aggregate was MIXED with both intervals individually recoverable in 30. Finally, the derived asset stayed MIXED and trusted in 30 of 30 derivations, with 0 essence mismatches. This is the object the vocabulary-versus-reduction distinction describes: the component histories are all present in the signed graph, and the aggregate interval claim exists only because the EM assertion computes it. Nothing in the validation pipeline required it to be conservative. A representative manifest is frozen as a fixture in the repository.

7.10. The cost of conservatism: claim dilution

Zero promotion is only half of a useful result, because safety alone is cheap: declaring the whole asset as every sample's dependency is safe under Proposition 2 and gives up all local specificity. We therefore measure what the declared footprints cost. The *dilution fraction* of a derived output is the fraction of its samples whose footprint-aware evidence record is strictly weaker, in claim, channel or support value, than the record computed over the nominal represented content. It measures conservatism added relative to the nominal map, not a comparison with a known true codec dependency.

The implementation widens footprints at the pull-backs of each source boundary shifted by $\pm$ the footprint, so dilution is confined to footprint-wide bands at ancestry boundaries; the per-sample oracle agrees with it on all 10 860 cases, including a battery of source intervals wider than every footprint.

In Table 6 and Figure 6, 4 of 8 transformations dilute nothing, lossless transcoding among them, and overlay because its nominal required-source set already contains every covering source. The remainder dilute the footprint-wide neighbourhood of each ancestry boundary. Median dilution is 0.86% for resampling, 20.24% for MP3, 31.34% for retained-run selection and 22.64% for time stretching. The single-transformation worst case is 58.33%. Each band has a *fixed radius in samples* and the clips average 2.85 s, so the same bands on a one-hour asset would occupy a fraction three orders of magnitude smaller: the fraction shrinks with duration, the absolute band does not.

The whole-asset comparator, one claim over the output computed from every source interval of the asset, promotes nowhere (0 promoted samples). Its median dilution is at least 99.99% on every corpus transformation except overlay, whose nominal map is already partly mixed (96.99%), against a median of at most 31.34% under the declared footprints; dilution is counted on provenance and support, so a sample whose nominal record already equals the whole-asset record is not diluted; this lowers overlay's value. A worked case shows why the other values approach 100%. In a clip of

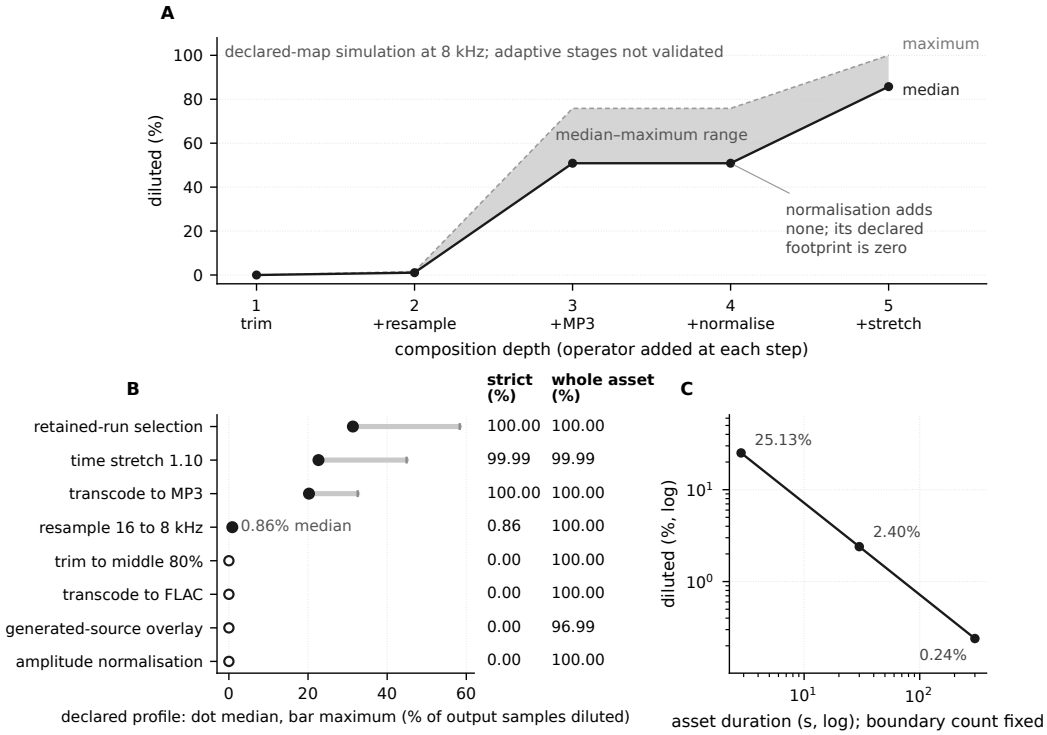


Figure 6: What the declared footprints cost, as the fraction of output samples whose claim is weaker than over the nominal map. (A) Through a composition chain, with the operator each step adds; the shaded band runs from median to maximum over clips and is not an uncertainty interval. Normalisation adds nothing: its content footprint is zero. (B) Each transformation alone over 600 clips: circles are medians under the declared profile and bars reach the maximum; the two columns give the strict-profile median and the whole-asset comparator's median, both near 100% for the measured operators. (C) The trim–normalise–MP3 chain at three durations with the boundary count fixed, on logarithmic axes. Dilution weakens a claim and never strengthens one.

captured, generated and captured intervals, the nominal record of a captured sample is captured with the captured support value, and of a generated sample generated with the lower generated one; the whole-asset record is MIXED with the lower value, the minimum. Every captured sample loses provenance and support, and every generated sample loses provenance, so every sample is diluted unless the clip carries one kind only. Three policies should be kept apart here. Boundary-only inheritance is specific and wrong on known counterexamples. The measured local footprints are specific and safe only where their containment holds, which the holdout of Section 7.8 refutes for the encoder and the stretcher. The strict profile is safe for the measured operators by construction and pays for it: median dilution is 100.00% for MP3, 99.99% for time stretching and 100.00% for retained-run selection, against 0.00% for lossless transcoding, whose footprint is analytical. Whole-asset fallback is thus the conservative reference: a local declaration is worth carrying when it keeps more specificity than that reference while its containment is justified.

Dilution grows with composition. Along the chain trim, resample 16 to 8 kHz, transcode to MP3, normalise, time stretch by 1.10, deliberately front-loading a rate change so that the large bands are scaled by it, median dilution reaches 85.74% by depth 5 with a worst case of 99.99%; depth 4 equals depth 3 to the last digit because the operator added there, normalisation, has a zero footprint. Whether a rate change rescales upstream bands is not isolated by this design, whose only rate change occurs before any upstream band exists. The chain is a model of how declared footprints

Table 6: Claim dilution: the fraction of output samples whose footprint-aware evidence is strictly weaker than the evidence over the nominal represented content, per transformation over 600 clips and along a composition chain over 200 clips. This is the price of the declared footprints, which combine analytical support with implementation margin, when the nominal map is exact.

Transformation	Clips	Median dilution	Max dilution	Clips affected
amplitude normalisation	600	0.00%	0.00%	0
overlay a generated source	600	0.00%	0.00%	0
resample 16 to 8 kHz	600	0.86%	1.39%	600
retained-run selection	600	31.34%	58.33%	600
time stretch 1.10	600	22.64%	44.92%	600
transcode to FLAC	600	0.00%	0.00%	0
transcode to MP3	600	20.24%	32.56%	600
trim to the middle 80%	600	0.00%	0.00%	0
composition depth 1	200	0.00%	0.00%	n/a
composition depth 2	200	1.06%	1.59%	n/a
composition depth 3	200	50.87%	75.84%	n/a
composition depth 4	200	50.87%	75.84%	n/a
composition depth 5	200	85.74%	99.99%	n/a
trim–normalise–MP3 chain, 2.85 s asset	1	25.13%	n/a	n/a
trim–normalise–MP3 chain, 30 s asset	1	2.40%	n/a	n/a
trim–normalise–MP3 chain, 300 s asset	1	0.24%	n/a	n/a

compound: its MP3 and stretch stages run at 8 kHz, configurations Section 7.7 did not probe, so it does not show that those declarations contain the chained audio's dependency. Holding a routine three-operator chain fixed (trim, normalise, MP3) and varying only duration, dilution is 25.13% at the corpus mean length, 2.40% at 30 s and 0.240% at five minutes: each tenfold increase in duration divides the fraction by ten, as fixed-width bands predict when the number of boundaries is held fixed, which it is here and would not be for an arbitrary chain. Dilution therefore dominates only where a deep chain of large-footprint operators meets a short asset with many ancestry boundaries, the case this measurement constructs; the quantity a deployment should carry is the band radius of Table 3, not a percentage measured on 2.85-second clips. This is the price of declaring conservative bounds instead of knowing exact codec dependencies, and it can be bought back without losing safety: any tighter map that remains an over-approximation reduces dilution and, by Proposition 2, cannot introduce promotion.

7.11. Consumer verification

A consumer holds the derived asset's assertion and, through the ingredient graph, each source's own interval evidence. With the dependency declaration of Supplementary Note S4, the verifier (`em_audio/consumer.py`) rebuilds the map, checks that it covers the output exactly and recomputes every interval's record. Three checks are reported separately. For declaration consistency, over the 4 800 corpus outputs of Experiment D, honest assertions verified in 4 800 of 4 800 under the declared profile and 4 800 under the strict profile; the decoded-length check and the signed workflow follow below. Boundary-only claims under the same declaration were flagged in 3 600 of the 3 600 promoting runs, and in 874 further outputs whose whole-output claim does not promote: 600 of them are the interval-level provenance promotions of Section 7.4, and the rest are other evidence-contract violations, a support value or lineage the sources do not give. One interval promoted after the fact was flagged in 2 804 of 2 804 cases, one dropped lineage entry in 3 000 of 3 000, and a declaration

missing its last piece in 4 200 of 4 200. A producer that declares zero footprints and computes its claims to match passed in 2 400 of 2 400: the check establishes consistency with the signed declaration, not that the declaration describes the processing, which needs other evidence: calibration probes, analysis of the algorithm, or an execution record.

The consumer must also take the output length from the audio rather than from the declaration. Given the decoded length of every Experiment D output, it rejected 1 223 declared-profile assertions whose modelled output length differs from the decoded one: in 888 the decoded output is longer than modelled and in 335 it is shorter. It verified 4 800 of 4 800 strict-profile assertions, whose maps are fitted. A declaration shortened consistently, map and length together, over unchanged audio was rejected in 4 800 of 4 800 cases.

Two further checks reduce reliance on shared implementation code and test the complete signed workflow. A separately written JavaScript verifier (`oracle_js/verify.js`), which imports nothing from the Python package and recomputes every output sample by brute force, returned the same verdict as the Python verifier on all 2 400 cases it was given, spanning every verdict class, with 0 disagreements. And over 20 clips the whole workflow ran through signed files: a source signed with its own interval assertion, a trim by stock FFmpeg signed twice with the source as `parentOf` ingredient, once honestly and once with one interval promoted. The verifier read the assertion from the derived asset's manifest, the source evidence from the ingredient's manifest and the length by decoding. Both versions validated as signed in all 20 cases; the honest one verified in 20 and the promoted one was flagged in 20, which is the difference between a valid signature and a consistent claim. End-to-end signed verification is thus demonstrated on 20 trim derivations; the larger evaluations above, over every transformation and both profiles, ran on the assertions as JSON.

The declaration raises the median assertion in this cohort, the corpus outputs of Experiment D, from 2 093 to 2 584 B, and recomputation took a median 0.16 ms per output. These figures come from a different cohort than Section 7.13, whose 1 333 B is an evidence-only assertion. The bookkeeping percentage reported there covers propagating intervals and building the assertion; decoding, signing, validation and consumer verification are not included in it.

7.12. Two-language differential

The Python reference implementation works with interval algebra over pulled-back source boundaries; the Node.js oracle evaluates evidence per output sample by brute force and run-length-encodes the result, with no interval algebra at all. The frozen corpus spans footprints wider than every source interval and footprints far narrower, the regime added after the dilution experiment (Section 7.10). It is stored as `fixtures/oracle_cases.jsonl`, one case per line carrying the interval model and the source evidence in a form that names neither implementation, and the oracle reads it as a separate process. On 10 860 cases the two agreed on interval structure, provenance state, lineage set and channel availability with 0 disagreements, and the maximum absolute difference in any support value was 0. Both implementations are author-written, so this is a differential test against transcription and interval-arithmetic error, not evidence of independence from the author (Section 9).

7.13. Cost

Complete-source bookkeeping cost a median 1.35 ms per audio-minute against 0.41 ms for boundary-only, a ratio of 3.34× computed from the unrounded medians, and 0.152% of the FFmpeg time it accompanies (Table 8). Signing took a median 16.9 ms per asset (IQR 16.2–17.2) and validation 14.0 ms (IQR 13.6–14.4). Manifest overhead was 14 932 B per asset, of which the EM

assertion was 1 333 B; the fixed per-asset component (certificate chain, COSE structure, hard binding) dominates, and the variable part grows linearly in emitted evidence intervals at 333 B and 21.0 μs per interval, approximately linear over the measured range up to 256. Cost therefore scales with how often ancestry changes along the timeline, not with duration. It also scales with how far the codec smears that ancestry: the assertion is 1 333 B in `wav` and 1 334 in `flac` against 2 114 in `mp3`, whose footprint adds cuts at each source boundary shifted by $\pm$ the footprint. An implementation that wants a ceiling on the count may coarsen the partition, merging adjacent intervals and emitting the meet over the merged range: each sample then carries the union of the merged required source sets, which by Proposition 2 can only weaken its claim, at the dilution cost of Section 7.10. Coarsening is therefore a conservative way to bound metadata size; we have not measured the trade-off between size and dilution it offers. The byte figure is deterministic and reproduces exactly; the microsecond figure is wall clock on one interpreter and one machine.

Both arms are timed in the same process, so load and clock changes move them together. Over 9 independent processes the absolute cost varied by 2.14% and the ratio by 2.05% (Supplementary Note S10); between machines the absolute figure differed by 20% and the ratio by 5.9% (Supplementary Note S8). The milliseconds show that the bookkeeping is cheap; they are not reproducible constants.

7.14. Independent reproduction

D. A. Balderrama-Alvarez (Universidad de Sonora), who is not an author and took no part in the design, ran the package for tag v1.0.0 on his own Linux machine from a package without result files (Table 10). The comparison criteria were fixed in code beforehand (Supplementary Note S7). This is independence of execution, not of implementation. His raw outputs are released unmodified in `results/independent/`, and the packaging history of the run is in Supplementary Note S8.

What reproduced. Of the 13 result files carrying a deterministic claim, 9 reproduced exactly on an operating system, processor architecture, Python version and FFmpeg major version that share nothing with the reference machine. Every conformance, promotion and oracle outcome is among them, including the 80 cases the superseded rule failed (Section 7.1), and decoded-essence identity before and after signing held for all 120 round-trips.

What did not. 216 deterministic differences remain. 3 are the size of a conformance suite that grew after his run; the other 213 trace to two declarations his FFmpeg build does not satisfy. For the MP3 encoder the reach on his build is 4 317 against a declared 2 304, and 110 output samples fall outside the declared support, all in the transient context. That removes the guarantee for those samples; whether it promotes depends on whether the omitted source's evidence is weaker than that of the sources the map includes. For retained-run selection, the mapping deviation rises from 681 to 12 505 samples against a margin of 4 096, a systematic offset (median 9 105.5), while its containment arm found no sample outside the declared support. Of the 4 operators with a non-zero footprint, 3, all but MP3, reproduced their reach exactly.

Two later runs, on macOS (an earlier release with a 885 828-check suite) and on Windows (tag v1.0.7), show that the version string does not fix the calibration. The macOS run reproduced every deterministic output; the Windows run's 69 differences all lie in the MP3 kernel measurements and the time-stretch deviation (463 against 471 samples), and the declarations still contained every measurement (0 probes outside). Three of the four rows of Table 10 report FFmpeg 9.0.1, and the MP3 reach takes two values among them. Nordvik and Axelsson (2022) found the same for

Table 7: Main results. Deterministic fixture rates characterise the frozen fixture family and are a mechanism stress rate, not a population prevalence estimate.

Check	Cases	Boundary-only	Complete-source
A Exhaustive finite-state conformance			
Source words over $\{C, G, \perp\}$, length ≤ 8	9,840	n/a	n/a
Operator cases	98,385	n/a	n/a
Contract checks	984,213 checks, 0 failed		
Named regression tests	49 tests, 49 passed, 0 failed		
B Deterministic adversarial timelines (10 000 frozen fixtures, 64 intervals)			
Chain depth 1	10,000	9,455 (94.5%)	0
Chain depth 3	10,000	9,455 (94.5%)	0
Chain depth 5	10,000	9,455 (94.5%)	0
Lineage omissions (depth 1)	10,000	10,000	0
Unverified → verified (depth 1)	n/a	7,222	0
C–D Mixed-origin audio corpus, stock-FFmpeg processing			
Exact propagation of constructed interval evidence	600	n/a	600
Promotion counts are whole output / any interval.			
amplitude normalisation	600	600 / 600	0 / 0
overlay a generated source	600	0 / 600	0 / 0
resample 16 to 8 kHz	600	600 / 600	0 / 0
retained-run selection	600	0 / 0	0 / 0
time stretch 1.10	600	600 / 600	0 / 0
transcode to FLAC	600	600 / 600	0 / 0
transcode to MP3	600	600 / 600	0 / 0
trim to the middle 80%	600	600 / 600	0 / 0
Q PartialSpoof v1.2, third-party mixed-origin speech			
amplitude normalisation	600	594 / 594	0 / 0
overlay a generated source	600	541 / 598	0 / 0
resample 16 to 8 kHz	600	594 / 594	0 / 0
retained-run selection	600	169 / 264	0 / 0
time stretch 1.10	600	594 / 594	0 / 0
transcode to FLAC	600	594 / 594	0 / 0
transcode to MP3	600	594 / 594	0 / 0
trim to the middle 80%	600	326 / 326	0 / 0
R PartialEdit v1.1, neural speech editing of genuine recordings			
amplitude normalisation	600	599 / 599	0 / 0
overlay a generated source	600	0 / 600	0 / 0
resample 16 to 8 kHz	600	599 / 599	0 / 0
retained-run selection	600	188 / 193	0 / 0
time stretch 1.10	600	599 / 599	0 / 0
transcode to FLAC	600	599 / 599	0 / 0
transcode to MP3	600	599 / 599	0 / 0
trim to the middle 80%	600	527 / 527	0 / 0
H Two-language differential oracle			
Frozen cases	10,860	n/a	0 disagreements

Table 8: Cost, as medians over 30 repetitions on one machine. These are measurements of a Python reference implementation, not language-independent constants.

Quantity	Median	IQR
EM bookkeeping	1.354 ms / audio-minute	1.332–1.457 ms / audio-minute
Boundary-only bookkeeping	0.406 ms / audio-minute	0.399–0.412 ms / audio-minute
EM / boundary-only ratio	3.34×	n/a
EM as a fraction of FFmpeg time (propagation and assertion construction only)	0.152%	n/a
C2PA signing	16.9 ms / asset	16.2–17.2 ms
C2PA validation	14.0 ms / asset	13.6–14.4 ms
Manifest size overhead	14,932 B / asset	14,932–14,934 B
EM assertion, evidence only (this cohort)	1,333 B / asset	1,333–1,334 B
EM assertion marginal cost	333 B / evidence interval	n/a
Assertion, Experiment D outputs, without declaration	2,093 B	n/a
Same, with dependency declaration	2,584 B	n/a
Consumer recomputation	0.16 ms / output	n/a

file-system implementations sharing one specification, and Külper et al. (2026) found forensic outputs drifting across software versions. We report the two Linux findings as findings and do not widen the declarations to pass them: a footprint table is a calibration for a stated build, and a new build needs the two-pass calibration tool run again.

8. Discussion

What is established. A complete-source rule can be implemented over media time against stock FFmpeg, on our corpus and on two third-party corpora. Three results should be kept apart. Under the declared profile the claims conform to the declaration on every fixture and clip, which is algebraic conformance relative to that declaration, not validation against the real dependencies. The holdout shows that those declarations can miss real dependencies, and that the miss produces promotion. The strict profile replaces every measured declaration by whole-asset dependency and promoted on none, which follows from Proposition 2. The contract travels in an existing signed standard without altering the audio, and its evidence bookkeeping costs a fraction of a percent of the processing it accompanies. The algebraic results are independent of processing build; implementation conformance and calibrated footprints require verification on the evaluated configuration (Table 9).

Relation to the standard and to adjacent work. The layer addressed here is the reduction over a vocabulary that already preserves what the problem needs, and it requires no specification change (Section 5.4). Detectors, watermarks and chunk-localised bindings (Ono, 2026) provide complementary evidence; this contract controls how recorded source evidence propagates through derivation. Without source evidence, the contract cannot establish captured or generated provenance. For the security of the specification itself we refer to a public analysis (Golaszewski et al., 2026).

What a consumer can check. Promotion can be checked only if the assertion carries the map, so the schema carries a dependency declaration (Supplementary Note S4). With it a consumer recomputes

Table 9: Claim–evidence boundary. Each row names a class of claim, the evidence used for it, what that evidence does not establish, and whether the independent reproduction of Section 7.14 reproduced it. Evidence from one row is never used to authorise a claim in another.

Claim class	Evidence used here	Not established by it	Independent reproduction
Operator conformance	Propositions 1, 2, Theorem 1; enumeration; scope battery; two-language differential	That any deployed system conforms	Yes: 0 failures; the 984 213-check suite of v1.0.7 reproduced on Windows
Constructed fixtures	10 000 adversarial timelines with a control arm	Any population rate or named product	Yes: 9 455 against 0, exactly
Realistic audio	600 constructed, 600 PartialSpoof and 600 PartialEdit clips, stock FFmpeg	Balanced coverage of speakers, codecs or edits	Yes for our corpus; the other two added later
Interval-map fidelity	Predicted versus decoded sample counts, against declared margins	Containment; margins on other builds	**No**: selection deviation 12 505 against a 4 096 margin
Kernel-support containment	280 probes in 4 contexts over 7 configurations; holdout challenge	Containment at untested inputs or builds	**No**: MP3 reach 4 317 against 2 304
Signed transport	120 round-trips over `wav`, `flac` and `mp3`; decoded-PCM identity	C2PA security; perceptual or cross-decoder equivalence	Yes on both machines
Consumer check	Recomputation from the declaration on three corpora, against decoded length; JavaScript verifier; signed workflow	That the declaration describes the processing; a non-C2PA consumer	Not yet: added later
Runtime cost	30 repetitions on one machine	Compiled or broadcast-scale cost	**No** for ms; ratio within 5.9%
Conservatism cost	Per-sample dilution against exact ground truth	Dilution against a codec's true dependencies	Yes, exactly

Table 10: The reference machine and the three third-party runs. *MP3 reach* is the measured kernel reach in source samples against a declared footprint of 2 304. The reference and two of the runs use FFmpeg 9.0.1; the reference and Windows builds of that one version do not agree, which is why the declaration is stated per build. Full environments and per-operator detail are Supplementary Note S8.

Run	Platform	FFmpeg	Verifier outcome	MP3 reach
Reference (this paper)	macOS 26.5.2	9.0.1	n/a	1 555
D. A. Balderrama-Alvarez	Linux 6.6.87.2	8.0.1-3ubuntu2	`REPRODUCTION INCOMPLETE`; 216 deterministic differences	4 317
B. C. Guerra Flores	macOS 26.3.1	9.0.1	Exit zero; 17 differences, all timings	1 555
D. A. Balderrama-Alvarez	Windows 11	9.0.1-full_build-www.gyan.dev	69 deterministic differences	1 541

every claim from the declaration (Section 7.11), demonstrated end to end through signed files on 20 trims. No consumer check reaches a declaration that misstates the operation or its footprint, whether through a stale calibration, a different build or intent; the dependency miss of Section 7.14 is of that kind, and the assertion alone cannot tell these causes apart.

Deployment and reading. The contract deploys as a library beside the processing step, whose caller already knows the parameters that fix the interval map, so nothing touches the signal path. An interval labelled MIXED is a statement about the declared sources, not about the audio, and a manifest without an evidence assertion says only that its producer did not implement the contract. An examiner reporting an interval should state its footprint basis and build from the declaration: an analytical or whole-asset basis carries the guarantee, a measured basis only at the operating points its calibration tested.

Beyond audio. Nothing in Section 4 refers to audio. The contract applies to other derived data whenever their required-source sets can be specified: temporal filtering of measured and imputed observations, image inpainting, robotic maps and LiDAR products that combine measurements with predictions, or video composed from captured and generated regions. Each needs domain-specific evidence channels and justified spatial or temporal dependency maps, and none is evaluated here. Preserving the distinction between observed, inferred and generated content does not by itself establish geometric accuracy, estimator confidence or operational safety.

9. Limitations

Independence. The oracle and the evidence policies are author-written; the differential tests transcription and interval-arithmetic error, not author independence. All processing is stock FFmpeg and all signing the official `c2patool`. The third-party runs of Section 7.14 are independent in execution only, against criteria fixed in code beforehand (Supplementary Note S7). A third-party reimplementation remains outstanding; it would run against the 10 860 frozen cases of `fixtures/oracle_cases.jsonl`.

One validator, not an interoperability result. Every round-trip is produced and checked by the official `c2patool` under a locally generated trust anchor, so *trusted* means trusted under that anchor. A second API, `c2pa-python`, also recovered the assertion, but it binds the same Rust core. No consumer implemented by a third party has read the assertion, so semantic interoperability is not established.

Corpus. The constructed corpus and the two third-party subsets test the mechanism across selected ancestry patterns; they do not provide balanced coverage of speakers, languages, recording conditions or editing methods, and support no rate claim about audio in general. The propagation rule never reads the audio; the footprint declarations of adaptive operators depend on the signal. Codec configuration, signal context and ancestry geometry are therefore the axes that can change the outcome, and more codec configurations are the direction Section 7.14 shows to matter.

Footprints are declarations. The analytical footprints hold for the exact configurations used here and the measured ones only at the tested operating points (Section 7.8). A different build, resampler setting, codec or tempo factor requires re-declaration with both passes of `tools/calibrate_footprint.py`, mapping and support, which exits non-zero when a declaration fails to contain its measured reach; until then the strict profile applies. Proposition 2 guarantees only that erring large is safe.

Content and control dependency. The contract tracks represented-content ancestry rather than all computational dependencies. In peak normalisation each output sample represents one source sample, while the gain that scales it depends on the whole signal, as the holdout challenge measures; input-dependent gain, encoder bit allocation and stretch alignment therefore fall outside this model. A deployment requiring all computational dependencies can assign whole-asset dependency to these operators, at the cost of local specificity.

Replay, identity and truth. A microphone recording a loudspeaker playing synthetic speech produces audio that is genuinely captured at the sensor layer; every capture-provenance system has this limitation, and the contract would propagate a captured claim, correct at the layer it models. Version 1 stays at the digital-asset and source-lineage level and detects no re-recording; capture mechanism, generation ancestry and speaker identity would need separate channels. A captured recording can also contain a lie, and the contract says nothing about semantic content.

From a machine-readable claim to an examiner's reading. The motivation drawn from Andersen et al. (2025) is a human-factors result: examiners erred on how a tool presented data. Whether an examiner reads MIXED correctly and acts on it is untested here and needs a study with examiners.

10. Conclusion

A representation-only transformation may change, compress, aggregate or reorganise content, but it must not silently increase the evidential authority of the output relative to the complete source material it represents. In audio that principle has an exact form: provenance is the union over the complete required source set, support the minimum, applicability the intersection, lineage the union, and an unverified source interval is absorbing. The algebra is not new. This work provides the executable temporal contract, the kernel-footprint treatment that makes it implementable against FFmpeg, a signed transport inside an existing standard, and a consumer that recomputes every claim. Complete-source propagation promoted in no interval on our corpus or on the two third-party corpora, leaves decoded WAV, FLAC and MP3 audio untouched, and its evidence bookkeeping costs a fraction of a percent of the processing it accompanies.

The guarantee is relative to the declared dependency map. The holdout challenge shows that the measured declarations for the encoder and the stretcher can miss real dependencies, and a miss produces promotion; the strict profile removes the dependence on them at the price of diluting essentially all output of those operators. Earlier independent runs reproduced the conformance, promotion and oracle outcomes of their release and found one footprint and one mapping margin that do not transfer to another build; the experiments added since have not been rerun independently. The contract cannot tell you whether speech is synthetic or truthful, and a valid signature proves only that an assertion was signed.

Data availability

The mixed-origin corpus is generated deterministically by committed code from LibriSpeech `dev-clean` (Panayotov et al., 2015), licensed CC BY 4.0, fetched by a checksum-pinned script, and from locally generated eSpeak NG output. The robustness arm also uses a Piper neural text-to-speech voice trained on the public-domain LJ Speech corpus, fetched and checksum-verified rather than committed, and pink noise generated locally, seeded per clip. Experiment Q uses 600 utterances

of the PartialSpoof v1.2 development set (Zhang et al., 2021b), licensed CC BY 4.0, fetched and checksum-verified by script rather than redistributed, and Experiment R 600 files of the PartialEdit v1.1 E1 subset (Zhang et al., 2026), CC BY 4.0, built on the VCTK corpus (Yamagishi et al., 2019), fetched the same way. No new human participants were recruited; existing public speech data were used, and no proprietary or login-walled data. The robustness arm uses a pretrained voice as distributed and trains or adapts none. Licences, versions, URLs, access dates, digests and redistribution clauses are recorded in `DATA_LICENSES.md`. Every result file, the frozen reference and every fixture are archived with the code in Zenodo, doi:10.5281/zenodo.23310554, which resolves to one immutable version; the identifiers of the selected PartialSpoof and PartialEdit utterances are in the archived result files.

Code availability

The reference implementation, the second-language oracle, every experiment, every frozen fixture and every machine-readable result are released under the MIT licence with pinned tool versions and one-command reproduction at `https://github.com/1D-Coda/em-audio`, public at submission rather than on acceptance. The unmodified outputs of the independent reproductions of Section 7.14, with their preflight reports, pipeline logs and classified comparisons, are released in `results/independent/`, `results/independent_mac/` and `results/independent_windows/`. The repository ships a frozen copy of the released results in `results/reference/`, the comparison target when `tools/verify_reproduction.py` runs from a downloaded archive, and a container image and continuous verification that re-measure the declarations on further builds, author-run evidence not reported here. The evaluated state is release tag v1.3.0, archived in Zenodo as doi:10.5281/zenodo.23310554, where the tag and the archive hold the same files.

CRediT authorship contribution statement

Alejandro Urias: Conceptualization, Methodology, Software, Formal analysis, Validation, Investigation, Visualization, Writing – original draft, Writing – review and editing.

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The author declares no competing financial or personal interests.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used Claude Opus 5 and Opus 5.5 (Anthropic) and ChatGPT (OpenAI) to improve language clarity and readability, and as programming assistants on the verification and reproducibility tooling. Every reported value was obtained by executing the committed experiments, and the author verified the code that produced it. After using these services the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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References

Andersen, D.B., Sunde, N., Porter, K., 2025. Tool induced biases? Misleading data presentation as a biasing source in digital forensic analysis. Forensic Science International: Digital Investigation 52, 301881. doi:10.1016/j.fsidi.2025.301881.

Burri, X., Casey, E., Bollé, T., Jaquet-Chiffelle, D.O., 2020. Chronological independently verifiable electronic chain of custody ledger using blockchain technology. Forensic Science International: Digital Investigation 33, 300976. doi:10.1016/j.fsidi.2020.300976.

Coalition for Content Provenance and Authenticity, 2026a. C2PA technical specification, version 2.4. https://spec.c2pa.org/specifications/specifications/2.4/specs/C2PA_Specification.html, accessed 19 August 2026.

Coalition for Content Provenance and Authenticity, 2026b. Use of Content Credentials to identify synthetic and non-synthetic content. C2PA implementation guidance. Published July 2026; https://c2pa.org/resources/, accessed 20 August 2026.

Denning, D.E., 1976. A lattice model of secure information flow. Communications of the ACM 19, 236–243. doi:10.1145/360051.360056.

European Parliament and Council of the European Union, 2024. Regulation (eu) 2024/1689 laying down harmonised rules on artificial intelligence (artificial intelligence act). Official Journal of the European Union, L series. URL: http://data.europa.eu/eli/reg/2024/1689/oj.

Fu, X., Wang, Z., Zhang, Y., Chen, Y., 2026. Trust the voice, hide the source: Anonymous provenance for verifiably edited audio. Cryptology ePrint Archive, Paper 2026/1308. https://eprint.iacr.org/2026/1308; preprint, not peer reviewed at the time of writing.

Gerhardt, M., Cuccovillo, L., Aichroth, P., 2024. Audio provenance analysis in heterogeneous media sets, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops, Workshop on Media Forensics, pp. 4387–4396. https://openaccess.thecvf.com/content/CVPR2024W/WMF/html/Gerhardt_Audio_Provenance_Analysis_in_Heterogeneous_Media_Sets_CVPRW_2024_paper.html, accessed 20 August 2026.

Goguen, J.A., Meseguer, J., 1982. Security policies and security models, in: 1982 IEEE Symposium on Security and Privacy, pp. 11–20. doi:10.1109/SP.1982.10014.

Golaszewski, E., Krawetz, N., Sherman, A.T., Zieglar, E., Matukumalli, S.K., Yus, R., Kegley, C.L., Barthel, M., Bowman, W., Barot, B., Kullman, K., 2026. Verifying provenance of digital media: Security analysis of C2PA and its implementation. Cryptology ePrint Archive, Paper 2026/804. https://eprint.iacr.org/2026/804; preprint, not peer reviewed at the time of writing.

Green, T.J., Karvounarakis, G., Tannen, V., 2007. Provenance semirings, in: Proceedings of the Twenty-Sixth ACM SIGMOD-SIGACT-SIGART Symposium on Principles of Database Systems (PODS '07), pp. 31–40. doi:10.1145/1265530.1265535.

Hargreaves, C., Nelson, A., Casey, E., 2024. An abstract model for digital forensic analysis tools – a foundation for systematic error mitigation analysis. Forensic Science International: Digital Investigation 48, 301679. doi:10.1016/j.fsidi.2023.301679.

Külper, M., Lambertz, M., Rybalka, M., 2026. Do you dare to try test-driven forensics? Increasing trust in desktop forensics with ADARE. Forensic Science International: Digital Investigation 57, 302115. doi:10.1016/j.fsidi.2026.302115.

Nemecek, A., He, H., Cheng, G., Ayday, E., 2026. Authenticated contradictions from desynchronized provenance and watermarking. arXiv:2603.02378. https://arxiv.org/abs/2603.02378; preprint, not peer reviewed at the time of writing.

Nordvik, R., Axelsson, S., 2022. It is about time–do exFAT implementations handle timestamps correctly? Forensic Science International: Digital Investigation 42–43, 301476. doi:10.1016/j.fsidi.2022.301476.

Ono, T., 2026. MerkleSpeech: Public-key verifiable, chunk-localised speech provenance via perceptual fingerprints and Merkle commitments. arXiv:2602.10166. https://arxiv.org/abs/2602.10166, submitted 10 February 2026; preprint, not peer reviewed at the time of writing.

Panayotov, V., Chen, G., Povey, D., Khudanpur, S., 2015. LibriSpeech: An ASR corpus based on public domain audio books, in: 2015 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pp. 5206–5210. doi:10.1109/ICASSP.2015.7178964.

Pham, L., Lam, P., Tran, D., Tang, H., Nguyen, T., Schindler, A., Skopik, F., Polonsky, A., Vu, C., 2025. A comprehensive survey with critical analysis for deepfake speech detection. Computer Science Review 57, 100757. doi:10.1016/j.cosrev.2025.100757.

Qamhan, M., Alotaibi, Y.A., Selouani, S.A., 2023. Transformer for authenticating the source microphone in digital audio forensics. Forensic Science International: Digital Investigation 45, 301539. doi:10.1016/j.fsidi.2023.301539.

San Roman, R., Fernandez, P., Défossez, A., Furon, T., Tran, T., Elsahar, H., 2024. Proactive detection of voice cloning with localized watermarking, in: Proceedings of the 41st International Conference on Machine Learning (ICML), pp. 43180–43196. URL: https://proceedings.mlr.press/v235/san-roman24a.html. peer-reviewed conference version; preprint arXiv:2401.17264.

Urias, A., 2026. EMTRF: Evidence-monotone terrain representation framework. Zenodo, software, version 1.1.3. doi:10.5281/zenodo.22143198.

Wang, X., Delgado, H., Tak, H., Jung, J.w., Shim, H.j., Todisco, M., Kukanov, I., Liu, X., Sahidullah, M., Kinnunen, T., Evans, N., Lee, K.A., Yamagishi, J., et al., 2026. ASVspoof 5: Design, collection and validation of resources for spoofing, deepfake, and adversarial attack detection using crowdsourced speech. Computer Speech and Language 95, 101825. doi:10.1016/j.csl.2025.101825.

Wang, X., Yamagishi, J., Todisco, M., Delgado, H., Nautsch, A., Evans, N., et al., 2020. ASVspoof 2019: A large-scale public database of synthesized, converted and replayed speech. Computer Speech & Language 64, 101114. doi:10.1016/j.csl.2020.101114.

World Wide Web Consortium, 2013. PROV-DM: The PROV data model. W3C Recommendation. https://www.w3.org/TR/prov-dm/, accessed 19 August 2026.

Yamagishi, J., Veaux, C., MacDonald, K., 2019. CSTR VCTK corpus: English multi-speaker corpus for CSTR voice cloning toolkit (version 0.92). University of Edinburgh, Centre for Speech Technology Research. doi:10.7488/ds/2645.

Zhang, L., Wang, X., Cooper, E., Evans, N., Yamagishi, J., 2023. The PartialSpoof database and countermeasures for the detection of short fake speech segments embedded in an utterance. IEEE/ACM Transactions on Audio, Speech, and Language Processing 31, 813–825. doi:10.1109/TASLP.2022.3233236.

Zhang, L., Wang, X., Cooper, E., Yamagishi, J., Patino, J., Evans, N., 2021a. An initial investigation for detecting partially spoofed audio, in: Proc. Interspeech 2021, pp. 4264–4268. doi:10.21437/Interspeech.2021-738.

Zhang, L., Wang, X., Cooper, E., Yamagishi, J., Patino, J., Evans, N., 2021b. PartialSpoof database, version 1.2. Zenodo. doi:10.5281/zenodo.5766198.

Zhang, Y., Tian, B., Zhang, L., Duan, Z., 2025. PartialEdit: Identifying partial deepfakes in the era of neural speech editing, in: Proc. Interspeech 2025, pp. 5353–5357. doi:10.21437/Interspeech.2025-942.

Zhang, Y., Tian, B., Zhang, L., Duan, Z., 2026. PartialEdit dataset, version 1.1. Zenodo. doi:10.5281/zenodo.18829689.